



Historical Perspective

Microscale characterization of abiotic surfaces and prediction of their biofouling/anti-biofouling potential using the AFM colloidal probe technique

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ABSTRACT

Given its detrimental consequences, biofouling is a central problem in industrial applications that affects submerged artificial abiotic surfaces. For instance, it shortens the service life of surfaces operating in marine and industrial environments, such as ship hulls, pipelines, heat exchangers, water filtration membranes, biosensors, and medical implants and concomitantly increases maintenance demands. In recent years, research has been propelled by the urge to minimize the replacement and maintenance costs related to the biofouling of abiotic surfaces. Colloidal probe atomic force microscopy (AFM) is a crucial analytical technique for quantifying interfacial interactions and unveiling the underlying biofouling mechanisms for a broad range of substrates. This technique is used to measure surface and colloidal forces by retracting individual colloidal particles immobilized at the apex of the AFM cantilever away from a planar surface under the conditions of choice. These forces are believed to dominate the biofouling behavior of surfaces. Several studies have utilized the colloidal probe AFM technique to estimate the adhesion of biofoulants to surfaces and elucidate the various aspects governing this process. This review summarizes the applications of colloidal probe AFM, including but not limited to characterizing the properties of abiotic surfaces, such as adhesive interactions, mechanical properties, and biofouling propensities. In addition, recent applications of colloidal probes relevant to biofouling are outlined. Furthermore, studies using various modified colloidal probes to test the biofouling resistance/anti-biofouling capacity of modulated abiotic surfaces are included.

1. Introduction

The atomic force microscope (AFM) is a powerful tool for characterizing the nanotopography of materials and their mechanical and chemical properties in different environments at a high resolution. AFM is non-optical and relies on a sharp tip that is mechanically scanned over a sample surface [1]. When the tip is placed close to the sample surface, it experiences attractive and repulsive interactions with the sample, causing cantilever deflection. This deflection is recorded as a function of position on the *xy* plane to obtain topographical images [2]. However, if the tip remains fixed and the cantilever is moved vertically toward and away from the sample in the *z*-direction, the deflection of the cantilever as a function of the piezo displacement can provide information regarding adhesive forces and mechanical properties of the sample, such as stiffness [2]. Such AFM force measurements are focused on probing the interfacial forces between surfaces across a medium [3]. However,

these measurements suffer from a major drawback: the unknown size and shape of the tip [4]. In 1991, Ducker et al. and Butt et al. independently addressed this limitation by introducing the colloidal probe technique [4,5]. In the colloidal probe technique, a spherical particle with a well-defined diameter (typically 2–20 μm) is glued or sintered to the free end of the cantilever, and the surface forces between this sphere and a flat surface are quantitatively measured [3]. Since colloidal particles are generally less than 1 μm in diameter, the nomenclature used for this technique (i.e., the colloidal probe) is, to some extent, a misnomer [6]. The interaction forces measured by the colloidal probe can be normalized by the known radius of the colloidal particle (F/R), which facilitates comparison between different measurements. The radius of the colloidal probe can be obtained via scanning electron microscopy (SEM) or AFM reverse imaging [6]. Although the colloidal probe technique has a lower spatial resolution, it has many advantages for quantitative force measurements, including a well-known geometry

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and size, higher applicability to theoretical models based on the sphere-plane geometry, higher sensitivity to weak forces, larger surface area available for coating or functionalization, less probability of contamination, wear and abrasion, and reduced stresses applied to sample surfaces [7].

Colloidal probe technology has multiple applications in research and industry fields [8]. It is mainly used to measure colloidal forces between homo- or hetero-particles or interfaces of various sizes and functionalities or to measure surface forces between a single colloidal solid or fluid material and a variety of substrates. Knowledge of such interaction forces is mandatory for understanding the stability of colloidal dispersions and the processes in which the adhesion phenomenon takes part or contributes most. Some examples of the fields that utilize the colloidal probe technique include mineral flotation [9,10], aggregation/agglomeration of suspended colloidal particles [11,12], drug delivery systems [13–15], tissue engineering [16–18], design of adhesives, glues, and coatings [19–21], and surface biofouling.

Biofouling can be defined as the undesired deposition, growth, and development of biofilms on abiotic surfaces [22–24]. In this study, we refer to abiotic surfaces as the artificial surfaces that are widely used in industrial applications and which are commonly exposed to biofouling. However, the attachment of microorganisms to abiotic surfaces is usually preceded by the adsorption of biomolecules (e.g., proteins, glycoproteins, lipids, nucleic acids, polysaccharides, humic substances, etc.) of various organic origins from the surrounding environments forming conditioning films. Once deposited on abiotic surfaces, these conditioning films can modify the physicochemical properties of surfaces (e.g., roughness, charge, hydrophilicity/hydrophobicity, chemical composition, etc.) and hence affect the initial bacterial adhesion [25–27]. When a cell succeeds in adhering to an abiotic surface or a conditioned abiotic surface, it starts to replicate and secrete extracellular polymeric substances (EPS) and form a three-dimensional biofilm matrix causing surface biofouling. These biofilms increase the mechanical stability of the embedded cells against cleaning processes and enhance their tolerance toward biocides [22,28]. Biofouling causes severe problems in industries that involve wetted or submerged abiotic surfaces [29]. For example, annual maintenance costs in the shipping sector are estimated to be as high as \$150 billion [30,31]. Biofoulant deposition on marine vessels increases hydrodynamic drag forces, resulting in increased fuel consumption and greenhouse gas emissions [32–34]. Similarly, contamination of food-contact surfaces with microbiological infectious agents causes foodborne diseases. The cost of foodborne illnesses in the United States has been estimated at \$152 billion per year [35]. Moreover, biofouling of filtration membranes presents a major obstacle to the feasible and widespread use of membrane based-water purification systems, which is urgently needed to overcome global water shortage problems [36]. Therefore, biofouling mitigation through modification of abiotic surfaces has become an active research field.

Several methods can be applied to assess and characterize surface biofouling. Some of these methods are based on characterizing the fouling layer, attached cells, or biofilm after their formation on the surface, while others real-time monitor the initial adsorption of biofoulants/cells to the surface, such as quartz crystal microbalance with dissipation (QCM-D) [37] and surface plasmon resonance (SPR) [38]. Characteristics of the formed fouling layer, such as its chemical composition, can be specified via spectroscopic techniques such as Fourier-transform infrared spectroscopy (FT-IR) [39] or energy dispersive x-ray spectrometry (EDX) [39]. In contrast, the topographical structure of the developed fouling layer/biofilm, as well as the number of viable cells attached to the surface, can be attained via microscopic techniques such as light microscopy [40], scanning electron microscopy [41], and confocal laser scanning microscopy [42]. Moreover, cell counting techniques such as the plate count method [43] can also be used to enumerate the viable cells on the surface. Although the spectroscopic and microscopic techniques provide valuable quantitative and

qualitative information on the fouling layer/biofilm, they may suffer from some drawbacks, such as the complexity of sample preparation, which may damage the fouling layer/biofilm, or the overlapping of fluorescence signals between auto-fluorescence biomolecules and fluorophores [44,45]. In comparison, QCM-D and SPR offer powerful, sensitive, easy-to-use, and label-free techniques to monitor fouling behavior. However, they require uniform and stable coating on the usually used gold-coated sensors, which may take prolonged times of optimization [46]. In addition, they show low sensitivity toward small molecules with low molecular weights [47,48]. Compared with these techniques, the colloidal probe technique has several advantages. It enables direct quantification of the interaction forces during all stages of fouling, starting from the initial attachment of the first molecules/cells on the clean surface and ending with the attachment of more molecules on the fouled surface. It does not require sample preparation and only needs a small piece of sample. Additionally, the colloidal probe can be functionalized (if needed) by a simple adsorption or coating process.

Remarkably, atomic force microscopy (AFM) is a sensitive and versatile tool for characterizing the surface properties of abiotic surfaces, such as surface roughness [49–51], surface heterogeneity [52,53], pore size distribution [54–57], nanomechanical properties [58,59], and interfacial interaction forces [58,60]. In particular, the colloidal probe technique can be used to understand the origin of the interfacial interactions between foulant molecules and surfaces, the intermolecular interactions between foulant molecules, and to reveal the underlying molecular mechanisms of the biofouling phenomenon. Furthermore, the approach and retraction cycles of the colloidal probe technique imitate the natural fouling process where biofoulants initially approach the surface, start experiencing long-range interfacial forces, and finally adhere to the surface due to short-range interactions. Therefore, the colloidal probe technique is crucial for quantifying and predicting the biofouling propensity of abiotic surfaces, and can support the efforts devoted to modifying these surfaces and enhancing their antifouling performance.

2. Calibration methods for the spring constant of AFM colloidal probe cantilevers

Prior to any AFM measurement, accurate determination of parameters such as the spring constant of the cantilever and the optical deflection sensitivity is necessary in order to convert the raw AFM data (photodiode reading versus piezo displacement) into force versus distance curves [3]. While there are several methods for calibrating the cantilever spring constant, such as the geometric [61,62], Cleveland [63], reference cantilever [64], Sader [65], thermal noise [66], and many other methods [67–69], no method universally applies. Each of these methods is based on a different technique that has certain advantages and limitations. For example, the *geometric method* calculates the spring constant based on the geometric dimensions and material characteristics of the cantilever. The spring constant k of V-shaped cantilevers can be written using the parallel beam approximation as follows:

$$k = \frac{Ewt^3}{2L^3} \quad (1)$$

where E is Young's modulus, and w , t , and L are the width, thickness, and length of the cantilever, respectively. The parallel beam approach approximates a V-shaped cantilever as two rectangular cantilevers lined up parallelly [62]. Since it is cubed in Eq. (1), a slight variation in the cantilever thickness can drastically alter the spring constant. In addition, Young's modulus of the cantilever may vary with the composition and the presence of oxide and gold coatings on its surface [3]. Therefore, the thickness and Young's modulus of the cantilever must be known accurately to calculate the spring constant. However, this requirement can be avoided using other methods. In the *Cleveland method*, the resonance

frequency ω is measured as a function of the mass M of an added end weight, and the spring constant is obtained from the slope of the linear plot of M versus $(2\pi\omega)^{-2}$ as follows [63]:

$$M = k(2\pi\omega)^{-2} - m^* \quad (2)$$

where m^* is the effective mass of the cantilever, which depends on cantilever geometry. In comparison, the *reference cantilever method* calibrates the cantilever spring constant by deflecting it against a pre-calibrated standard cantilever as follows [64]:

$$k_1 = k_0[C(D_z/N) - 1] \quad (3)$$

where k_1 is the unknown spring constant, k_0 is the reference spring constant, C is the nA-to-nm conversion factor, D_z is the z-displacement of the sample stage, N is the deflection of the unknown cantilever after contact (in nA), and (D_z/N) is the inverse of the slope of the force–distance curve in the contact regime.

Sader et al. initially proposed a method for calibrating the spring constant that relied on the dimensions, mass, and resonance frequency of the cantilever [70]. However, the cantilever thickness is generally measured using an electron microscope, which can be time-consuming, and the mass or density of commercial cantilevers is usually unknown and difficult to measure. Therefore, Sader et al. later proposed a different method (the *Sader method*) for calibrating rectangular cantilevers that only depended on the plan-view dimensions (length and width), resonance frequency, and quality factor of the cantilever in the fluid as follows [65]:

$$k = 0.1906\rho_f w^2 L Q_f \Pi_i(\omega_f) \omega_f^2 \quad (4)$$

where ρ_f is the density of the fluid, Q_f is the quality factor in fluid (typically air), and w and L are the width and length of the cantilever, respectively. Π_i is the imaginary part of the hydrodynamic function (Π), which depends only on the Reynold number ($Re = \rho_f \omega_f w^2 / 4\eta$), where η is the viscosity of the fluid and ω_f is the fundamental mode resonant frequency. However, this method is only valid for rectangular cantilevers with an aspect ratio (L/w) exceeding 3, and when $Q_f \gg 1$, which is typically applied when the cantilever is in air.

The *thermal noise method* is one of the most popular methods for calibrating the spring constants of cantilevers. It is implemented in many commercial AFM software packages and does not require additional equipment. First proposed by Hutter et al. [66], the thermal noise method relates the thermal fluctuations of the cantilever to its spring constant via the equipartition theory, which states that, at thermal equilibrium, every quadratic term in the total energy of a system equals $\frac{1}{2}k_B T$, where k_B is the Boltzmann constant and T is the absolute temperature [71]. If the cantilever is bent in the z-direction, its potential energy is $\frac{1}{2}kz^2$. Thus, according to the equipartition theory, $\frac{1}{2}k_B T = \frac{1}{2}k\langle\Delta z^2\rangle$, where $\langle\Delta z^2\rangle$ is the mean square deflection of the cantilever due to thermal vibrations [71]. As such, the amplitude of the thermal noise of a cantilever with a free end can be expressed as [66]:

$$k = \frac{k_B T}{\langle\Delta z^2\rangle} \quad (5)$$

The background noise of the instrument can be subtracted by analyzing the thermal fluctuations in the frequency domain through thermal power spectrum analysis (the square of the amplitude of thermal fluctuations of the cantilever as a function of frequency). None of the noise sources resonates at the resonant frequency of the cantilever, and can therefore be easily filtered out [66]. The remaining resonant peak is fit with a Lorentzian function, and because the integral of the power spectrum P_v (in V^2) is the mean square of the cantilever fluctuations in the time domain, the spring constant can be calculated as follows:

$$k = \frac{k_B T}{P_v} \quad (6)$$

When the thermal noise method is used to calibrate the spring constant, several corrections are usually introduced to account for various factors, such as the contribution of a single vibration mode (to account for deviation from the simple harmonic model, a factor of 0.971 is used for rectangular cantilevers and a factor of 0.965 for V-shaped cantilevers) [71], difference in the inclination to the deflection conversion ratio of a free cantilever compared with that of a supported cantilever (0.817 for rectangular and 0.764 for V-shaped cantilevers) [71,72], tilt of the cantilever [73–75], and laser spot size and position [76]. These factors can be inserted into Eq. (6), to give [77,78]:

$$k = \frac{\alpha_i k_B T}{(\hat{s} S_z)^2 P_{v,i}} \quad (7)$$

$$k = \frac{\alpha_i k_B T}{\chi_i^2 (\hat{s} S_z^*)^2 P_{v,i}} \quad (8)$$

where α_i represents the fraction of the total oscillation energy stored in the i^{th} vibration mode (for the first mode of a rectangular tipless cantilever $\alpha_1 = 0.971$), χ_i is the conversion factor of the sensitivity from the end-loaded shape to the freely oscillating shape of the cantilever (≈ 1.09 for rectangular cantilevers), S_z is the deflection sensitivity, S_z^* is the deflection sensitivity that converts the voltage output to the effective deflection (the effective deflection is the deflection read from the instrument after determining the sensitivity from the contact part of a force curve on a hard surface) [3], $P_{v,i}$ is the area under the resonant peak for the i^{th} mode, and $\hat{s}$ accounts for the condition of loading on the cantilever (i.e., the tilt of the cantilever, the torque applied on the cantilever due to the finite height of the tip, and the loading point of the cantilever, which is the center of the tip for standard cantilevers and the end of the cantilever for tipless cantilevers) as given by [73–75]:

$$\hat{s}^{-2} = \left(\frac{L - \Delta L}{L} \right)^2 \left[\frac{1 - \frac{3}{2} \frac{H/L}{(L - \Delta L)/L} \tan\theta}{1 - 2 \frac{H/L}{(L - \Delta L)/L} \tan\theta} \right]^2 \cos^2(\theta) \quad (9)$$

where ΔL is the tip loading offset, H is the tip height and θ is the cantilever tilt angle. Combining α_i and χ_i^2 in Eq. (8) into one factor, β_i , we obtain:

$$k = \frac{\beta_i k_B T}{(\hat{s} S_z^*)^2 P_{v,i}} \quad (10)$$

At this point, two questions arise: whether the above techniques are valid for calibrating colloidal probe cantilevers after colloidal particles have been attached to their ends and whether the spring constants of tipless cantilevers differ from those of colloidal probe cantilevers. First, for colloidal probe cantilevers, both the spring constant and optical deflection sensitivity vary with different colloidal loading points along the cantilever axis as follows [79]:

$$k_{L-\Delta L} = \left(\frac{L}{L - \Delta L} \right)^3 k_L \quad (11)$$

$$S_{L-\Delta L} = \left(\frac{L - \Delta L}{L} \right) S_L \quad (12)$$

where ΔL is the loading position offset defined as the distance between the end of the cantilever and the center of the colloidal particle, $k_{L-\Delta L}$ and k_L are the offset-and end-loaded spring constant values, respectively. Similarly, $S_{L-\Delta L}$ and S_L are the offset-and end-loaded deflection sensitivity values, respectively. Notably, the maximum loading point depends on the diameter of the colloidal particle (Fig. 1). Thus, to ensure that the colloidal particle (the cell in the case of cell probes) is the first thing to touch the sample surface and to exclude the interactions between the cantilever and the sample, the maximum distance between the colloidal particle and the end of the cantilever should not exceed

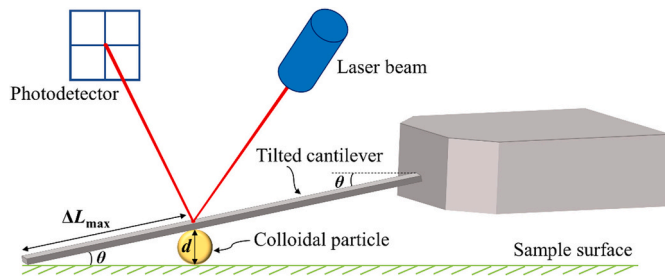


Fig. 1. Schematic illustration of the maximum loading point of a colloidal particle on a tilted cantilever.

$\Delta L_{\max} = \frac{d}{\sin\theta}$, which is the upper limit of the distance between the colloidal particle of diameter d and the free end of the cantilever [80].

Considering the cantilever tilt, the torque acting on the attached sphere of radius (R), and the loading point offset for a rectangular cantilever, the spring constant will be given by [70,75,78]:

$$k_{CP} = \left\{ \left[1 - \frac{3R/L}{2(L - \Delta L)/L} \tan\theta \right] \cos^2(\theta) \right\}^{-1} \left(\frac{L}{L - \Delta L} \right)^3 k_{TL} \quad (13)$$

where k_{CP} is the corrected spring constant of the colloidal probe and k_{TL} is the spring constant of the tipless cantilever.

We now summarize the approaches used in the literature to calibrate colloidal probe cantilevers. A common approach is to calibrate the spring constant before attaching the colloidal particle to the end of the cantilever using one of the aforementioned methods and ignoring any effects that the attached particle exerts on the cantilever stiffness. Craig et al. developed an in situ method that utilizes the hydrodynamic drag force exerted on the cantilever and colloidal particle to calibrate the spring constant of the cantilever [81]. Laurent et al. investigated the effect of colloidal mass loading on the eigenmode shapes and resonance frequencies of AFM cantilevers using a differential interferometer that detects deflections as differences in the optical path of two laser beams reflected off the cantilever's chip and the cantilever. Their findings demonstrated that gluing a large bead (10% of the length of a soft cantilever, $k \approx 0.1$ N/m) could change the first eigenmode slightly. Meanwhile, the higher oscillation modes changed considerably, and their nodes shifted toward the free end of the cantilever. In addition, heavier attached masses resulted in a more significant contribution from the first mode and weaker contributions from the higher modes. Therefore, the traditional thermal noise method, which measures the spring constant from the first flexural eigenmode, could still be applied to calibrate the colloidal probe cantilever [82]. Heim et al. showed that the equipartition theory which relates the Brownian motion of the cantilever to its spring constant, and forms the foundation of the thermal noise method, applies to colloidal probe cantilevers as well. They measured the spring constant using the thermal noise method and found that the resonance frequencies of the first and higher modes shifted to lower values after a particle is fixed to the end of a cantilever. The spring constants calculated for bare and colloidal probe cantilevers varied on average by approximately 5%, 20%, and 50% for ideal rectangular cantilevers (with three-cornered free ends), flat-tipped V-shaped cantilevers, and for non-ideal cases (misplaced particles), respectively [83].

The thermal noise method is used extensively to calibrate the spring constants of cantilevers with an accuracy of $\pm 10\%$ and a precision of $\pm 5\%$ [84]. Most commercial AFM instruments have a thermal noise calibration routine built into their software. However, this method requires prior determination of the deflection or optical lever sensitivity, which makes it relatively less straightforward. Optical sensitivity is usually acquired by ramping the cantilever against a rigid surface that does not undergo significant deformation during force loading. If the compliance of the cantilever is much lower than that of the tip and the sample, the piezo displacement of the sample will be equal to the tip

deflection once they are in contact. Thus, the optical lever sensitivity simply equals the inverse slope of the linear part in the contact regime of a force–distance curve measured on a hard surface, i.e., the inverse optical lever sensitivity (InvOLS). However, this procedure is difficult to apply in certain cases, such as when a soft or functionalized colloidal probe cantilever is used. The reference cantilever method, wherein the cantilever is pressed against a reference cantilever with a known spring constant, is also problematic. Furthermore, friction upon scanning a colloidal probe against a rigid sample or reference cantilever may preclude the measurement of linear graphs due to stick-slip motion [85]. In contrast, the Cleveland method does not require previous knowledge of the deflection sensitivity; however, the uncertainty in measuring the added sphere diameter can cause a 15–30% error in the measured spring constant [86]. In addition, variations in the positioning of the mass spheres on the cantilever are a major source of error in the Cleveland method [70]. Furthermore, glue ($\sim 10^{-15}$ L) is typically used to attach the masses to the cantilevers [6], and, although the use of small amounts of glue to attach the spheres in the Cleveland method was found to have a negligible effect on the calibration curve [70], if not correctly handled, this might introduce additional errors to the calculated spring constants and deflection sensitivities.

The thermal noise method has also been applied in combination with other techniques to calibrate the spring constant of the cantilever and InvOLS in a contactless manner. Higgins et al. combined the Sader method, which enables the contactless measurement of the spring constant of rectangular cantilevers in air, with the thermal noise method to calculate the non-contact InvOLS [87]. A discrepancy was expected between the vibration mode of the free cantilever (thermally driven) and the bending of the supported cantilever (statically end-loaded). Therefore, the calculated values of the non-contact InvOLS were corrected to obtain contact InvOLS values using a factor of 1.09 (χ_i) [87], which was also proposed by Walters et al. [72] based on the deflection data from Butt et al. [71]. Attard et al. suggested a complete non-contact approach to calibrate optical sensitivity by relating the voltage noise of the photodiode to the thermal fluctuations of the cantilever angle [88]. The authors asserted that their analysis was accurate because they accounted for the ratio of the thermal fluctuations measured in the first vibration mode to the total vibration modes, tilt of the cantilever, torque of the extended tip, and friction effect with the substrate. However, to obtain the spring constant of the static cantilever from angular fluctuations using their method, a measurement of the slope of the constant compliance region of a force–distance curve recorded on a hard surface was still required [88]. Chung et al. used an electrostatics-based technique to determine the AFM force sensitivity. This method was coupled with the thermal noise spectrum of a colloidal probe cantilever to calibrate the non-contact InvOLS and spring constant [85]. In addition, empirical estimations of the correction factors were made between the static deflection sensitivity, which was determined using the constant compliance method, and the dynamic or non-contact deflection sensitivity, which was determined using a combination of electrostatic and thermal-based techniques. The correction factors obtained for the colloidal probes were found to be between 1.04 and 1.13 in air (with a mean of 1.07) and 1.02 in liquid. These correction factors accounted for the dynamic mode shape and the laser spot size and position along the cantilever. The authors pointed out the difficulty of obtaining a complete non-contact calibration method because such colloidal probe correction factors lack analytical expressions and can only be obtained experimentally by comparing the static (contact) and dynamic (non-contact) values of InvOLS [85].

Another technique that offers a non-contact thermal calibration method to determine the deflection sensitivity of the cantilever without the need to touch a surface is the laser Doppler vibrometry (LDV) [89]. The LDV uses the Doppler shift of a laser beam reflected from a moving object and enables deflection measurements with a higher signal-to-noise ratio. Excellent agreement (less than 5% uncertainty) was reported comparing the spring constants of tipless cantilevers determined

using LDV with those determined on a standard AFM system using either the thermal noise or Sader method, provided that the limitations of the methods and the instruments were accounted for [86]. Gates et al. used LDV to determine cantilever spring constants and experimentally calculated the mode correction factors (MCFs) required in the thermal calibration method from the areas under the first few resonance peaks of a flexural-type mode [90]. They showed that adding a mass to the end of a tipless cantilever decreases the experimentally determined MCFs and resonance frequencies. In addition, their results indicate that when proper correction factors are used, the static spring constants of the cantilevers are maintained after attaching the colloidal mass [90]. Although the LDV method is a competitive approach that has been successfully applied, it requires additional software and special equipment.

Implementing the thermal noise method for the calibration of cantilever spring constants requires the mass to be uniformly distributed along the cantilever. However, this may not be the case when large colloidal spheres with a radius and mass comparable to or larger than the length and mass of the cantilever, respectively, are attached to the end of the cantilever, decreasing the resonant frequency and increasing the quality factor and amplitude of the first resonant peak [78]. The deflection sensitivity for such large probes can be estimated by the method of Chung et al., which uses the mean slopes of the approach and retraction curve [85], or using the non-contact approaches. Chighizola et al. proposed an alternative method to calibrate colloidal probe cantilevers by calibrating the tipless cantilever before attaching the mass with the thermal noise method and measuring the deflection sensitivity using the standard contact approach or by using other deflection sensitivity-free calibration methods such as the LDV or Sader methods [78]. After attaching the mass, the loading position can be measured, and the spring constant can be corrected using Eq. (13). The deflection sensitivity can then be calculated from Eq. (8).

If the thermal noise and reference cantilever methods are used to calibrate large colloidal probe cantilevers directly [74], the spring constant should be corrected as follows [78]:

$$k_{CP,c} = \frac{\delta_i}{\delta_{i,0}} k_{CP,0} \quad (14)$$

where $k_{CP,c}$ and $k_{CP,0}$ are the corrected colloidal probe spring constant and that obtained from the thermal noise method, and δ_i and $\delta_{i,0}$ are the correction factor of the colloidal probe and the standard pre-factor in the AFM software, respectively.

2.1. Calibration methods for the spring constant of soft AFM colloidal probes

Determining the optical lever sensitivity from a force–distance curve can be problematic when probing soft samples or when using soft colloidal probes. In the first case, InvOLS is usually determined at the beginning of the measurement on a hard sample. The hard sample is then removed, and a soft sample is placed on the sample stage. For most AFMs, switching samples requires removing the cantilever holder from the AFM head and placing it back again. This may change the cantilever configuration and laser spot characteristics and, consequently, change the InvOLS. In the second case, if an external micromanipulation system is used to attach the soft colloid, the tipless cantilever needs to be removed from the AFM head after being calibrated so that the colloid can be attached. This will also change the InvOLS. Furthermore, the standard procedure for measuring InvOLS cannot be used for soft colloidal probes, because the soft tip deforms while being pressed against a rigid substrate. However, any error in measuring the optical lever sensitivity is doubly reflected in the spring constant, as indicated in Eq. (7). Therefore, it is necessary to accurately determine InvOLS and various procedures have been proposed in the literature to adjust the InvOLS accordingly.

In a recent study, inconsistencies in Young's modulus measurements of soft and biological samples between different AFMs were linked to uncertainties in determining the deflection sensitivity and spring constants of the cantilever [91]. Therefore, the authors proposed an inverse procedure to precisely measure Young's moduli for soft and biological samples. This procedure, denoted as SNAP, involved determining the cantilever's spring constant using an accurate method, such as a vibrometer, followed by using the thermal noise method to adjust the InvOLS until the exact value of the spring constant was obtained [91]. Other studies determined InvOLS for colloidal elastomer probes such as poly(dimethylsiloxane) from the slope of the contact region of a force–distance curve while pressing the apex of the cantilever against a steep edge [92,93]. The non-contact approach, which is similar to the SNAP procedure, has also been used to determine InvOLS for soft colloidal probes. In this approach, cantilevers with known spring constants are calibrated using the thermal noise method by adjusting the input value of InvOLS until the value of the spring constant matches its known value [93]. To prepare cell probes, such as bacterial probes, it was suggested that soft biological cells are attached to hard colloidal probes after measuring the InvOLS on a hard surface. One measurement of the InvOLS is sufficient because the positioning of the cell on the colloidal probe is implemented using AFM in that procedure without requiring an external micromanipulation system, i.e., the cantilever configuration is preserved [94]. A procedure based on the use of an external micromanipulation system to prepare cell probes indicated that focusing the laser spot at the same position used during cantilever calibration and obtaining a sum value similar to that obtained before results in a negligible change in the InvOLS [80]. In contrast, Webber et al. estimated the spring constant of a droplet-loaded cantilever by modeling the deformable interfaces of two oil droplets: one anchored to the cantilever and the other immobilized on the substrate. Their model accounted for the influences of droplet deformations, colloidal surface forces, and hydrodynamic interactions [95]. Interested readers are referred to the following cited references for further details on the measurements and modeling of surface interactions that involve bubble/drop deformable interfaces [96–104].

3. Primary surface and intermolecular interactions that govern surface biofouling

Since biofouling can be most simply defined as the adsorption or adhesion of biofoulants/cells to solid surfaces immersed in water, it is generally described by the classical Derjaguin–Landau–Verwey–Overbeek (DLVO) theory [105,106]. The DLVO theory is based on an energy balance between two types of interactions: the electrostatic double layer interactions (EL) and the electrodynamic or Lifshitz–van der Waals interactions (LW), as follows:

$$\Delta G_{132}^{TOT(DLVO)} = \Delta G_{132}^{EL} + \Delta G_{132}^{LW} \quad (15)$$

where $\Delta G_{132}^{TOT(DLVO)}$, ΔG_{132}^{EL} and ΔG_{132}^{LW} are the total free energy of interaction, the electrostatic energy of interaction, and the Lifshitz–van der Waals energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium) respectively. The EL interactions result when charged surfaces with a net electrical surface potential are placed in an electrolyte. They decay exponentially as a function of distance and the thickness of the diffuse double layer (i.e., Debye length, $1/\kappa$), which is strongly influenced by the ionic strength of the medium. In comparison, the LW interactions comprise the van der Waals–London (dispersion or induced dipole-induced dipole), van der Waals–Debye (induction or dipole-induced dipole), and van der Waals–Keesom (orientation or dipole-dipole) forces. The LW interaction energies decay in the non-retarded range with distance as l^{-2} for a parallel-plate geometry and l^{-1} for a two-spheres geometry [107].

The DLVO theory alone usually fails to explain the biofouling

behavior of surfaces [108–110]. Therefore, van Oss et al. [111] suggested an extended-DLVO form (the XDLVO theory), which could predict the adhesion/biofouling of surfaces more accurately. The XDLVO theory accounts for, in addition to the EL and LW interactions, the polar electron-acceptor/electron-donor or Lewis acid–base interactions (AB), as follows:

$$\Delta G_{132}^{\text{TOT(XDLVO)}} = \Delta G_{132}^{\text{EL}} + \Delta G_{132}^{\text{LW}} + \Delta G_{132}^{\text{AB}} \quad (16)$$

where $\Delta G_{132}^{\text{AB}}$ is the Lewis acid–base energy of interaction between entities 1 and 2 immersed in liquid medium 3. For the adhesion or biofouling to be favored, $\Delta G_{132}^{\text{TOT}}$ must be negative. Lewis acids and/or bases, have two distinct (AB) properties that can vary independently: electron acceptivity and electron donicity (usually expressed in terms of surface tensions γ^{AB} = $2\sqrt{\gamma^+ \gamma^-}$). These two different polar AB properties allow the molecules or particles to interact with each other both in cohesion and in adhesion modes [112]. However, if only one single polar parameter exists (i.e., a monopolar surface as will be discussed below), it cannot contribute to its energy of cohesion. In other words, the polar energy of cohesion will arise between the electron acceptor of material 1 and the electron donor of material 1; the same for material 2; while the polar energy of adhesion will arise between the electron acceptor of material 1 and the electron donor of material 2; and between the electron donor of material 1 and the electron acceptor of material 2. If materials 1 and 2 are immersed in liquid medium 3, analogous AB interactions will arise between materials 1 and 2 with liquid 3. The net of all of these interactions can be positive or negative. The AB interactions decay exponentially with distance according to a decay length of $\lambda \approx 0.6\text{--}1$ nm, which corresponds to the radius of gyration of the commonly occurring small clusters of hydrogen-bonded water molecules and which does not change significantly with the ionic strength of the medium. The AB interactions prevail in aqueous systems and can be as important or even more important than the EL and LW interactions [112]. For example, at ionic strengths ≥ 0.1 M, the decay rate for the AB interactions will be less steep than that of the EL interactions, given that the decay length (λ) of water is 1 nm which is also the value of the Debye length ($1/\kappa$) at 0.1 M for 1–1 electrolytes [112].

The AB interactions comprise many subsets, such as hydrogen-donor/hydrogen-acceptor or Brønsted interactions, hydrogen bonding interactions, π electrons interactions, steric interactions, hydrophobic interactions, and hydration interactions (hydrophilic repulsion). The hydrophobic interactions or the hydrophobic effect can be defined as the tendency of apolar molecules or particles to aggregate when immersed in water. According to van Oss, the sole driving force, in most cases, for hydrophobic interactions between apolar molecules or molecules of relatively low polarity is the strong polar (AB) energy of cohesion between the surrounding water molecules (due to hydrogen bonding) rather than the van der Waals attractions between hydrophobic molecules or particles. It should be noted that only one of the interacting surfaces should be hydrophobic for a hydrophobic interaction to occur. A closely linked interaction to hydrophobic interactions which is its repulsive counterpart is the hydration interactions or the hydration pressure. These two interactions, either in the attractive (hydrophobic interactions) or in the repulsive mode (hydration interactions) are both of polar (AB) origins [113]. As most solid polar surfaces and many hydrophilic biopolymers (or cells covered by biopolymers) possess a high electron donor parameter (γ^-) and a small or negligible electron acceptor parameter (γ^+) and thus they are known as monopolar surfaces, they usually repel each other when immersed in water or other polar liquids [114,115]. Most monopolar polymers and biopolymers have an apolar surface tension component of $\gamma^{\text{LW}} \approx 40$ mJ/m², as such their electron donor parameter (γ^-) should be equal to or larger than 28.3 mJ/m² for them to repel each other in water. Many monopolar polymers and biopolymers fulfil that condition and so are very soluble in water. Van Oss et al. attributed the repulsion observed between uncharged

surfaces in water, which was originally described as hydration pressure, to the polar repulsive forces between monopolar surfaces immersed in water, rather than to the structural forces connected with water or with water of hydration [114]. However, as that repulsion was noticeable at distances from a few to several water molecule diameters, van Oss et al. demonstrated that it is due to a manifestation of the water molecules orientation in the first hydration layers [114]. Moreover, the hydrophilic repulsion will only prevail when it is stronger than the always present hydrophobic and van der Waals attraction energies [116].

In Eq. (16), $\Delta G_{132}^{\text{EL}}$ (if applicable) is determined by electrokinetic measurements. However, the EL interactions at contact distances play a minor role compared to the contributions of the LW and AB interactions and hence are usually neglected [117–119]. The interfacial part (IF) of the interaction energy, i.e., ($\Delta G_{132,0}^{\text{LW}} + \Delta G_{132,0}^{\text{AB}}$) at contact distances ($h_0 = 0.157$ nm) is determined by [112]:

$$\Delta G_{132}^{\text{IF}} = \Delta G_{132,0}^{\text{LW}} + \Delta G_{132,0}^{\text{AB}} = \gamma_{12} - \gamma_{13} - \gamma_{23} = 2 \left(\sqrt{\gamma_3^{\text{LW}}} - \sqrt{\gamma_1^{\text{LW}}} \right) \left(\sqrt{\gamma_2^{\text{LW}}} - \sqrt{\gamma_3^{\text{LW}}} \right) + 2 \left[\sqrt{\gamma_3^+} (\sqrt{\gamma_1^-} + \sqrt{\gamma_2^-} - \sqrt{\gamma_3^-}) + \sqrt{\gamma_3^-} (\sqrt{\gamma_1^+} + \sqrt{\gamma_2^+} - \sqrt{\gamma_3^+}) - \sqrt{\gamma_1^+ \gamma_2^-} - \sqrt{\gamma_1^- \gamma_2^+} \right] \quad (17)$$

where γ_{ij} is the interfacial surface tension between materials i and j , γ_i^{LW} , γ_i^+ and γ_i^- are the apolar Lifshitz–van der Waals, polar Lewis–acid and polar Lewis–base components of the surface tension, respectively. In general, $\Delta G_{132}^{\text{IF}}$ can be either positive or negative, as both $\Delta G_{132,0}^{\text{LW}}$ or $\Delta G_{132,0}^{\text{AB}}$ can independently be positive or negative.

4. Applications of colloidal probe AFM in the characterization of abiotic surfaces and more general applications relevant to biofouling

Since its development, the colloidal probe technique has been used in several applications. The following sections summarize applications of the colloidal probe technique to characterize abiotic surfaces along with an overview of the latest general applications of the colloidal probe technique in relevance to biofouling.

4.1. Characterization of the properties of abiotic surfaces

Although imaging of surfaces with colloidal probes suffers from limited resolution and significant loss of nanoscale details, the colloidal probe technique is capable of characterizing the adhesion and mechanics of abiotic surfaces with sufficient spatial resolution. Compared with standard AFM sharp tips, colloidal probes are significantly larger, which gives rise to more averaged values of surface properties with lower standard deviations and improved sensitivity to weak interactions. In addition, most of the microorganisms implicated in the biofouling of surfaces, including bacteria, yeast, algae, and diatoms, have dimensions similar to colloidal probes (2–20 μm). Hence, colloidal probes best simulate their actual adhesion to abiotic surfaces. Moreover, the mechanical stress and strain induced in the sample during indentation with colloidal probes are distributed over a larger contact area; therefore, the probe is less likely to puncture or disrupt the sample surface. Colloidal probes have a well-defined geometry that can be easily characterized and analyzed using theoretical approaches (such as DLVO and XDLVO) and mechanical contact models. In contrast, sharp tips are more susceptible to wear and abrasion, resulting in an indeterminate contact surface area. The following sections provide examples of abiotic surface properties that can be acquired using the colloidal probe technique.

4.1.1. Adhesive properties and biofouling potential of abiotic surfaces

Colloidal probe force measurements provide information on the fundamental interactions responsible for the attachment of biofoulants

to surfaces and predict its underlying mechanisms. Therefore, colloidal probes can be employed to quantify the biofouling potential of surfaces in terms of adhesion forces (attractive forces or repulsive electrostatic or steric forces) (Fig. 2), with various micrometer-sized colloidal probes, which can be measured as a function of multiple factors to facilitate the optimization of process and surface conditions [120,121]. For example, Bowen et al., who pioneered colloidal force measurements on filtration membranes [121], used coated colloidal and cell probe techniques (these techniques will be discussed further in the coming sections) to measure the adhesive forces between a protein (bovine serum albumin, BSA) and a yeast cell (*Saccharomyces cerevisiae*) and two different polymeric ultrafiltration (UF) membranes [122]. As agreed with the biofouling behavior of these membranes in industrial processes, their study demonstrated the potential of such techniques to assess membrane biofouling using only a small piece of the membrane. Although the viability of the yeast cell in their study was not inspected, the authors stated the importance of assessing the viability of the immobilized cells when performing such measurements. Wang et al. developed polyvinylidene fluoride (PVDF) membrane-coated and foulant-coated colloidal probes to measure foulant–membrane and foulant–foulant adhesion forces, respectively, between the foulants: (BSA, sodium alginate, humic acid (HA), secondary wastewater effluent organic matter), and a PVDF UF membrane [123]. The authors found that foulant–membrane adhesion forces were positively correlated with the rate of flux decline in the initial filtration stage. Conversely, foulant–foulant adhesion forces were related to the pseudo-stable flux and cake layer structure in the later filtration stage. Mo et al. studied the effect of the density of carboxylic acid groups on the membrane surface on the organic fouling of thin-film composite polyamide nanofiltration (TFC NF) membranes [124]. Two membranes were fabricated by reacting piperazine with either trimesoyl chloride (TMC) or isophthaloyl chloride (IPC), which has a lower carboxylic acid group content. Their observations confirmed the dominant role of calcium ions in bridging the carboxylic groups of the membrane and the foulant, hence in the TMC membrane's fouling. In contrast, the adhesion forces and rupture distances were statistically similar with and without calcium ions for the IPC membrane. Therefore, decreasing the density of the surface carboxylic acid groups of polyamide membranes manufactured with IPC monomers could inhibit calcium bridging with alginate and enhance the membrane's antifouling properties (Fig. 3). However, this improved antibiofouling performance of the IPC membrane is achieved at the expense of reduced water permeability. Therefore, the authors proposed combining TMC, IPC, and/or other monomers to optimize water

permeability. Zaouri et al. used colloidal probes coated with extracellular organic matter of *Skeletonema costatum* (a diatom) to study the metal oxide ceramic membranes fouling by the algogenic organic matter during episodic algal blooms [125]. Their results indicated that the attractive forces (jump-to-contact events) and maximum adhesion forces were higher for ZrO_2 surfaces than for Al_2O_3 , demonstrating that Al_2O_3 is more suitable as a ceramic membrane material for the filtration of hydrophilic compounds.

Ávila-Sierra et al. investigated the effect of surface characteristics, including chemical composition, surface roughness, and wall temperature of 316 L stainless-steel (SS316L), a common substrate in the food industry, on interfacial interactions [126]. Adhesion force measurements were performed in ambient air on SS316L substrates of different finishing grades using a borosilicate colloidal probe with a diameter of $\sim 6\ \mu\text{m}$. The adhesion was dominated by capillary force, which was controlled by the humidity of the environment, chemical composition, roughness, and modulus of the substrate. The adhesion force on the substrates with mirror and satin finishes was in the range of 1.5–2.5 μN , while the average adhesion force on the brush finish increased to 3.5 μN . The enhanced adhesion observed for the roughest finish (brush) was attributed to its higher contact surface area (Fig. 4A). More favorable adhesion on a whey protein concentrate (WPC)-fouled surface was attributed to the polar groups on the protein surface, as evidenced by contact angle measurements. The range of the adhesion forces measured between the borosilicate colloidal probe and the WPC foulant increased with increasing temperature, implying a heterogeneous foulant surface, and increased random molecular orientation (Fig. 4B). However, the effect of SS316L temperature on the adhesion measured between the borosilicate colloidal probe or WPC foulant and SS316L surface was not reported. When the temperature was increased, the total surface free energy of both the SS316L surface and the foulant layer which is divided to polar and disperse (apolar) components remained constant. In comparison, the polar part of the surface energy increased significantly by 3.4% with a temperature increase from 25 to 80 $^{\circ}\text{C}$, while the disperse part decreased slightly. In addition, Young's modulus of the WPC foulant increased with temperature: $3.3 \pm 1.3\ \text{GPa}$, $3.7 \pm 0.3\ \text{GPa}$, and $3.9 \pm 0.7\ \text{GPa}$ at 25, 50, and 75 $^{\circ}\text{C}$, possibly due to the configuration of protein molecules during deposition. Their results demonstrated the effect of the substrate temperature on the orientation of the molecules at the surface and their packing within the foulant layer.

Similarly, Horseman et al. investigated the role of surface roughness on the foulant–surface interactions using two different types of colloidal probes: a polyethylene (PE) colloidal probe to mimic the behavior of deformable hydrophobic colloidal foulants (e.g. oil droplets and proteins), and a polystyrene colloidal probe functionalized with carboxylic acid groups (C-PS) to simulate the fouling by organic matter that is typically rich in carboxylic acid groups (e.g. HA and algae) [127]. Rough surfaces were fabricated by coating quartz microscope slides with multiple layers of silica nanoparticles (SiNPs). Hydrophilic surfaces were prepared by grafting poly(ethylene glycol) (PEG) using silane chemistry, whereas hydrophobic surfaces were synthesized by surface grafting of fluoroalkylsilane. For hydrophilic surfaces, the maximum adhesion forces and rupture distances measured in a 10 mM aqueous solution of NaCl were considerably lower for rough surfaces compared to those measured for smooth surfaces (Fig. 5). The authors attributed this to the lower interfacial area between the probe and the rough surface than that with the smooth surface; i.e., there was less net adhesive energy. Furthermore, lower adhesion forces were measured with the C-PS colloidal probe than those with the PE colloidal probe. This was rationalized by the slightly larger and softer PE colloidal probe, which resulted in a larger interfacial area between the probe and the surface. Additionally, the PE probe is weakly charged, while the C-PS probe is highly negatively charged, which would lead to a high repulsive barrier between the C-PS probe and the surface. However, the probe hydrophilicity/hydrophobicity role in the adhesion forces with hydrophilic/hydrophobic surfaces while fixing all other factors, such as the probe

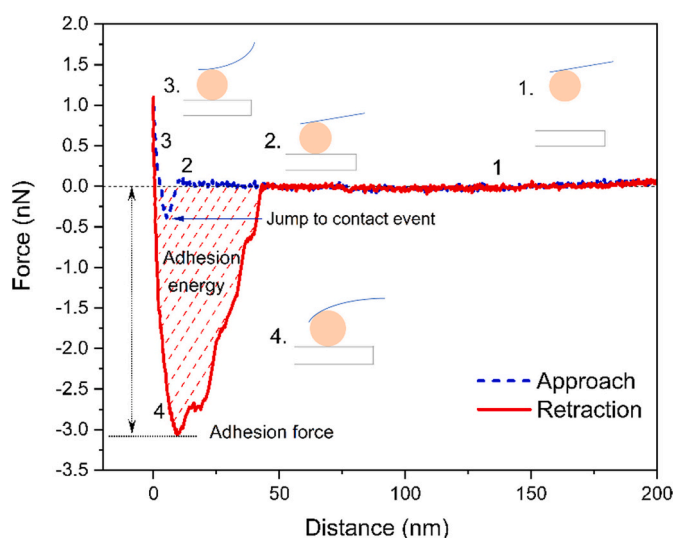


Fig. 2. A typical AFM force measurement between a colloidal probe and a sample surface.

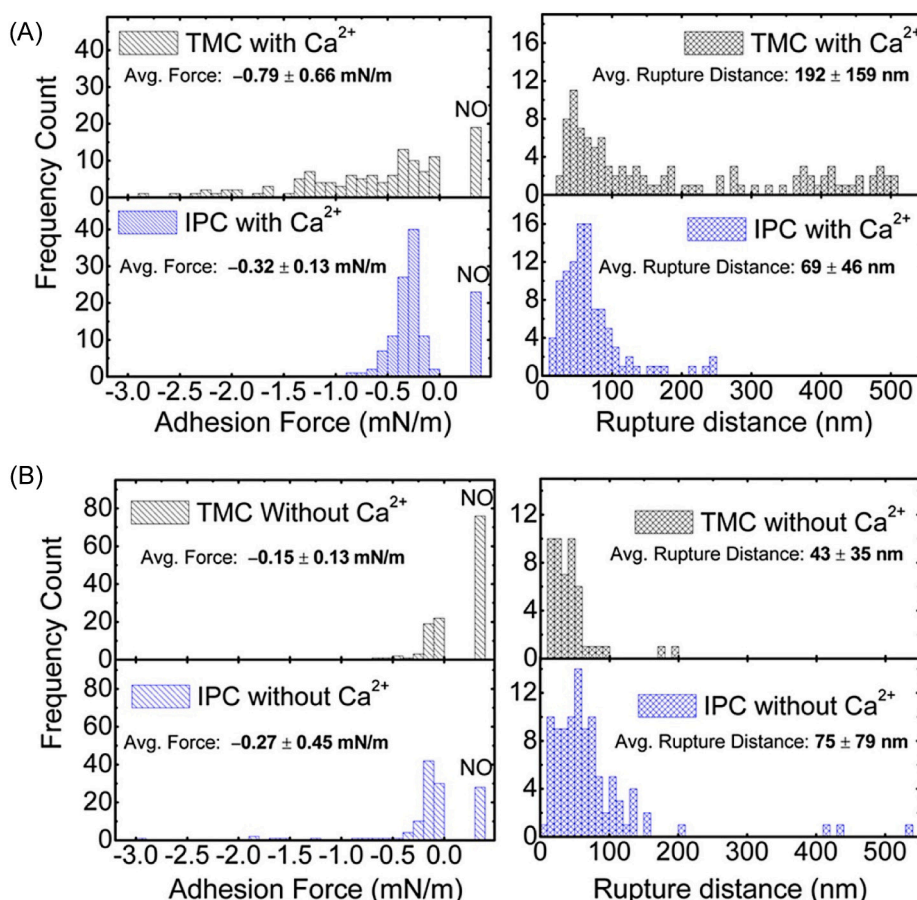


Fig. 3. Effect of surface carboxylic acid density of polyamide NF membranes (TMC: high carboxylic acid content, and IPC: low carboxylic acid content) on the adhesion forces and rupture distances between a carboxylate-modified latex colloidal particle used as a surrogate of alginate and the membranes (A) with Ca^{2+} and (B) without Ca^{2+} . Adapted with permission from [124].

charge, area, and stiffness, was not reported. Furthermore, the greater hydration of the C-PS probe rendered its contact with the hydrophilic surface energetically unfavorable because the net interfacial energy is increased. In contrast, the strong adhesive forces and large rupture distances observed for the rough hydrophobic surface, irrespective of the type of colloidal probe used, were explained by the capillary bridge interaction between the probes and the air film present on the rough hydrophobic surface in the Cassie-Baxter state. The strong long-range attractive interactions observed between the hydrophobic PE probe and the hydrophobic surfaces were attributable to hydrophobic-hydrophobic interactions. The authors verified that the fabrication of antifouling surfaces through modifying surface texture and roughness in order to mitigate organic fouling is based on the hydrophilicity/hydrophobicity of surfaces.

Siebrath et al. investigated the impact of water pretreatment on the organic fouling of reverse osmosis (RO) membranes using colloidal silica probes because they have the same charge as the pristine polyamide membranes, although it has a different hydrophobicity [128]. To optimize the pretreatment stage, adhesion forces were measured between silica probes and three polyamide surfaces fouled with the effluent organic matter (EfOM) originating from three wastewater treatments: conventional pretreatment or tertiary wastewater (TWW), conventional and biofiltration pretreatment (TWW + BF), and conventional and UF pretreatment (TWW + UF). The greater adhesion force was measured for the conventional TWW (7.72 ± 1.6 nN), followed by TWW + BF (1.62 ± 0.6 nN), then followed by TWW + UF (1.09 ± 0.5 nN). The lowest force of (0.06 ± 0.1 nN) was obtained for the pristine polyamide membrane. A similar trend was observed with QCM-D measurements. The authors ascribed these findings to the differences in the composition of the three

types of EfOM adsorbed to the polyamide surface. Therefore, the three types of EfOM were further fractionated, characterized and analyzed for their adsorption propensities using QCM-D. The effluent generated from (TWW + UF) contained less biopolymers, while the one generated from (TWW + BF) contained less hydrophobic compounds. Their QCM-D results identified the strongest adhesion for biopolymers, and the lowest adhesion for the lower molecular weight humics and building blocks. However, colloidal AFM measurements were not performed for the fractionated types of EfOM. The authors attributed the higher adhesion observed for biopolymers using the QCM-D to the hydrogen bonding between the hydroxyl groups of the biopolymer and the polyamide membrane, which are stronger than those between the carboxyl groups of humic substances and the membrane. Finally, the authors suggested assessing the fractional compositions of the fouled layers on the polyamide membrane compared with the fractions of EfOM solutions.

To untangle the fouling and defouling mechanisms of proteins, saccharides, and natural organic matter (NOM) on polymeric membranes, Fu et al. investigated the filtration properties of polyvinyl chloride (PVC) UF membranes using BSA, dextran (DEX), and HA as model foulants [129]. The fouling and defouling performances of the PVC membranes were studied by monitoring the flux decline and recovery during filtration and physical backwash, respectively. AFM measurements of foulant-foulant and foulant-membrane adhesion forces were performed using foulant-coated colloidal probes (BSA, DEX, and HA solutions). For all foulants used, the adhesion forces of the membrane-foulant were stronger than those of the foulant-foulant. The adhesion forces were greater for PVC-BSA (-3.8 nN) than PVC-HA (-0.03 nN), and the foulant-foulant interactions followed the order: BSA-BSA > HA-HA, as per

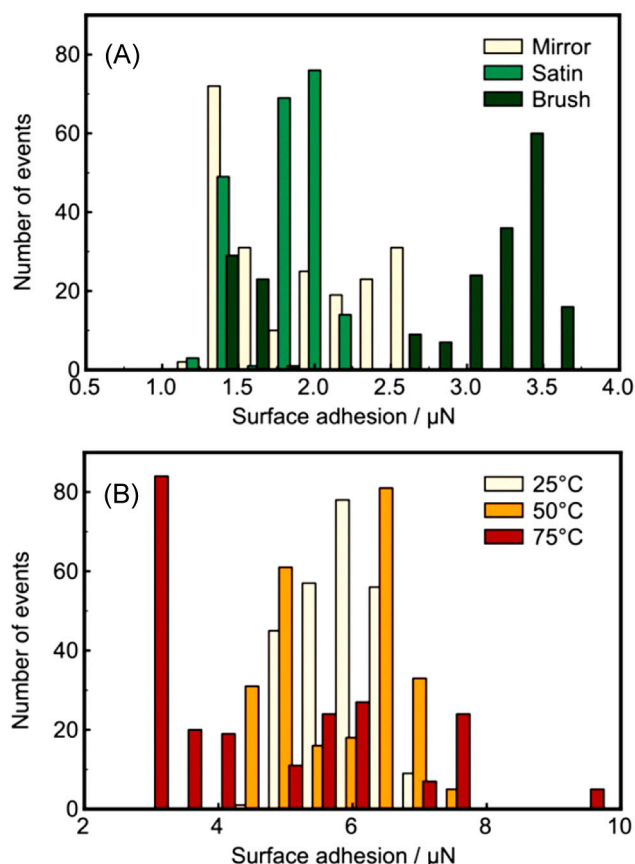


Fig. 4. Adhesion forces between the colloidal probe and (A) SS316L with different surface finishes, and (B) WPC-fouled SS316L surface under different surface temperatures. Adapted with permission from [126].

the calculated defouling rate order ($\text{BSA} < \text{HA}$). In addition, a positive correlation ($R^2 = 0.845$) was observed between the particle detachment rate (determined from the foulant–membrane adhesion force at the initial filtration stage, or the sum of the adhesion forces of the membrane–foulant and foulant–foulant at the later filtration stage) and defouling rate (flux recovery rate) for BSA and HA foulants that resulted in cake layer formation. However, because of its smaller molecular size, the DEX defouling rate did not follow this correlation because DEX partly clogged the membrane pores. The molecular and/or colloidal size of foulants is of importance when studying porous surfaces. This is because it controls the fouling mode whether it is cake layer mode or internal clogging of pores. Therefore, the authors demonstrated that their analysis could be further improved by accounting for factors such as foulant colloidal sizes, foulant layer coverage ratio, and foulant layer compressibility under the conditions of operation and the shear rate.

Powell et al. quantified the work of adhesion or the adhesion energies for virgin and fouled RO membranes represented by the area under the retraction curves using colloidal silica probes [130]. The shape of the force–distance curve described by the work of adhesion can provide more information on interaction mechanisms. The work of adhesion will be greater than the adhesive force (measured as the minimum of the retraction curves) if deformation occurs when the probe interacts with the sample surface or if multiple ligands contribute to the interactions because the probe has to peel away and break many bonds while retracting from the sample.

4.1.2. Mechanical and conformational properties of abiotic surfaces

Quantitative characterization of the mechanical properties of abiotic surfaces is critical, not only for the mechanical integrity and/or chemical stability of these surfaces but also for their role in the adhesion of

biofoulants such as bacteria. Earlier studies have pointed to the mechanoselectivity of bacteria, and several correlations between bacterial adhesion and substrate stiffness have been reported [131–133]. This implies that the mechanical properties of abiotic substrates can be used as indicators of biofouling propensity. The local micromechanical properties of abiotic surfaces can be extracted from indentation measurements using colloidal AFM probes. Furthermore, Young's modulus of the sample can be derived by analyzing the contact region of the force–distance curves using one of the theoretical models of continuum contact mechanics (Fig. 6A). Common models used to describe the elastic deformation of surfaces include the Hertz model, in which the probe-to-sample adhesion is neglected [134]; the Johnson–Kendall–Roberts model, which accounts for the adhesion within the probe-sample contact area [135]; and the Derjaguin–Müller–Toporov (DMT) model, which considers the adhesion outside the contact area [136]. For example, Powell et al. applied the Hertz model to derive Young's moduli from indentation measurements (using silica colloidal probes with a radius of 3 μm) of virgin and industrially fouled RO membranes in different electrolyte solutions [130]. The authors were able to use the micromechanical properties of the membranes to optimize the process conditions to achieve a higher mechanical robustness of the membrane surface and a lower mechanical strength of the fouling layer.

Unlike sharp AFM tips, the extensive surface area of colloidal probes markedly reduces the stress applied to the sample, fulfilling the requirement of elastic models of contact mechanics (i.e., falling within the linear elastic stress–strain region). The Hertz model applies to small deformations when the contact radius (a) is much smaller than the radius of curvature (R) of the colloidal sphere (i.e., $a \ll R$) [137]. The Hertz model was initially derived for the deformation of two elastic spheres [134]. The most commonly used form of the Hertz model describes the limiting case of a rigid sphere indenting an infinitely elastic half-sphere as follows:

$$F_{\text{paraboloid}} = \frac{4}{3} \frac{E}{(1 - \nu^2)} R^{\frac{1}{2}} \delta^{\frac{3}{2}} \quad (18)$$

where $F_{\text{paraboloid}}$ is the loading force of the paraboloid indenter, E is Young's modulus of the sample, δ is the indentation depth, and ν is the Poisson ratio of the sample. Another prevalent model is the Sneddon model, which is derived for large indentations of several geometries (e. g., spheres, paraboloids of revolution, cones, ellipsoids of revolution, and flat-ended cylinders) [138]. However, the Sneddon model for spherical geometry, which best describes the colloidal probe geometry, does not have an analytical solution in a closed form, and thus requires numerical solutions. Instead, it provides the applied force as a function of the contact radius as follows:

$$F_{\text{sphere}} = \frac{E}{2(1 - \nu^2)} \left[(a^2 + R^2) \ln \left(\frac{R + a}{R - a} \right) - 2aR \right] \quad (19)$$

The indentation depth (δ) is a function of a according to

$$\delta = \frac{1}{2} a \ln \left(\frac{R + a}{R - a} \right) \quad (20)$$

Eqs. (19) and (20) converge toward the Hertz model (Eq. (18)) in the limit of small deformations and large sphere radii (i.e., $a \ll R$) [139]. Notably, at low indentation depths, the parabolic and spherical indentation profiles are identical [140]. Therefore, the Hertz model can be viewed as a good approximation of the Sneddon model under small indentation depths or when using colloidal probes; however, this is not the case for sharp AFM tips [137].

Accurate determination of the contact point, where the tip starts to penetrate the sample, in the force–distance curve is essential for the correct and reliable interpretation of the data. In most studies, the contact point is defined as the point at which the deflection of the cantilever starts to deviate from the horizontal x-axis [5,130,141,142].

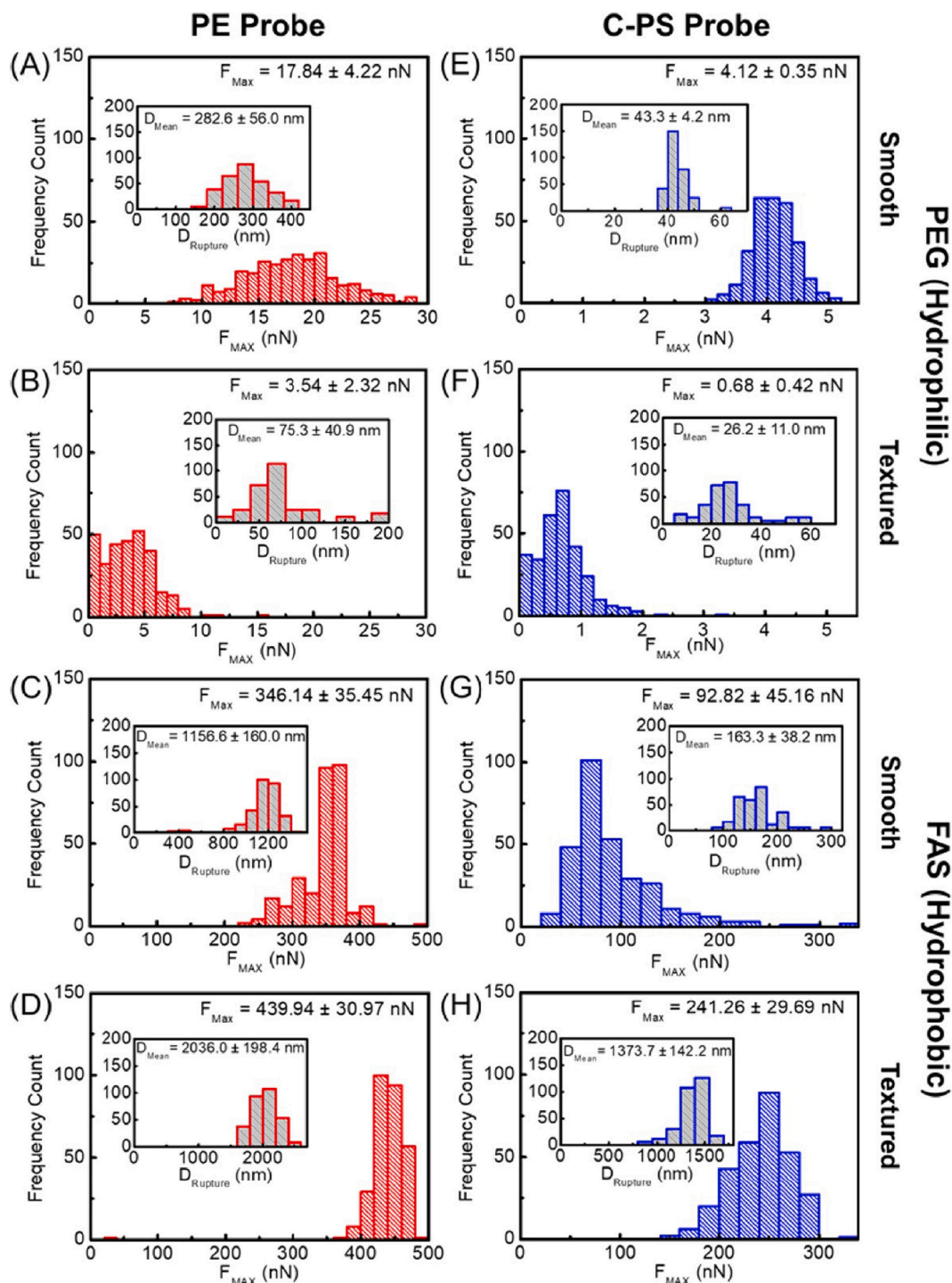


Fig. 5. Histograms of maximum adhesion force (F_{max}) and rupture distances (d_{Rupture} , insets) quantified at 300 locations on the sample surfaces. Adhesion statistics of PE colloidal probe interacting with the (A) smooth hydrophilic surface, (B) rough hydrophilic surface, (C) smooth hydrophobic surface, and (D) rough hydrophobic surface. Adhesion statistics of C-PS colloidal probe interacting with the (E) smooth hydrophilic surface, (F) rough hydrophilic surface, (G) smooth hydrophobic surface, and (H) rough hydrophobic surface. Reprinted with permission from [127].

Nevertheless, if linearized contact mechanics models are applied, the force–distance data can be fitted without identifying the exact location of the contact point [143]. Additionally, the effect of tip radius was investigated by Chyasnachyus et al. who found that the force–indentation curves acquired for the same material surface differed when probed with tips of different radii (they used 20 and 330 nm tips) [140]. Furthermore, they elucidated that the stress distribution below the tip–sample contact area extends further into the subsurface when

large-radii probes were used. Thus, larger probes can detect modulus values that match the bulk mechanical properties of the material rather than the surface properties [140]. Similarly, Dimitriadis et al. observed good agreement between Young's moduli of thick poly(vinyl alcohol) gels measured with AFM colloidal probes (2 and 5 μm) and a macroscopic measurement of the material itself [144]. In contrast, using standard sharp AFM tips resulted in higher estimates of Young's moduli values [144].

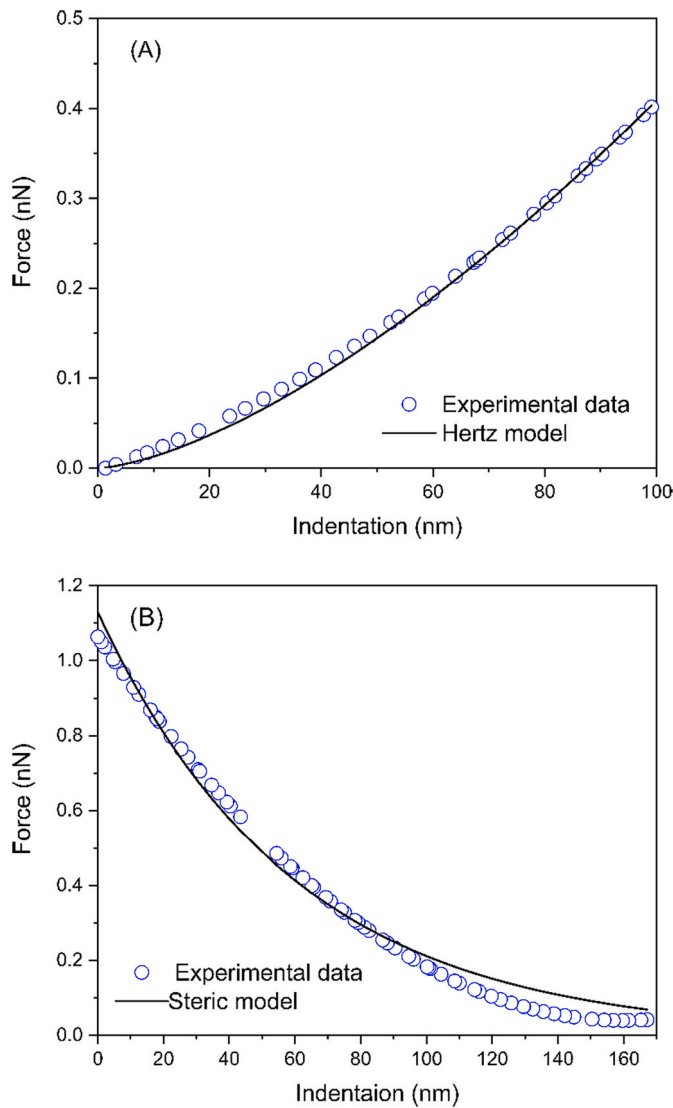


Fig. 6. (A) A representative approach force-indentation data obtained by a colloidal probe. The solid line represents the Hertz model fit (Eq. (18)) to the experimental data to find E . (B) A representative approach force-distance curve obtained by a colloidal probe. The solid line represents the steric model fit (Eq. (23)) to the experimental data to find L_0 and Γ .

The conformational and mechanical properties of the polymer brush coatings on abiotic surfaces, such as the grafting density, elasticity, and stiffness (apparent Young's modulus), have also been determined using AFM colloidal probe measurements (Fig. 6B). Recent studies suggested that the structural and mechanical properties of polymer brushes play an essential role in their biofouling potential. Although the relationships between the structural properties of polymer brushes, surface hydration, and biofouling resistance are not fully understood, several studies have highlighted the crucial role of optimizing the brush thickness and/or grafting density to ensure complete surface coverage and enhance surface resistance against protein biofouling [145–149].

Typical AFM tips have small radii in the range of 2–60 nm and can perturb the polymer brush only locally, while colloidal probes can approach the brush-coated surface by compressing the entire polymer brush [150]. In a study by Kutnyansky et al., a zwitterionic poly(sulfobetaine methacrylate) (PSBMA) brush was grafted onto a planar Si surface, and a poly(methyl methacrylate) (PMAA) brush was grafted onto a colloidal probe by surface-initiated atom transfer radical polymerization (SI-ATRP) [151]. The PMAA brush was modified with the

adhesion-promoting peptide sequence arginine-glycine-aspartic acid (RGD). Force-distance curves were obtained for systems where the polymer brush on the colloidal probe interacted with unmodified surfaces (asymmetric system) or modified surfaces (symmetric system). The grafting density of the polymer brush for each system was determined by applying a box-like model based on the Alexander-de Gennes theory [152]. An average grafting density was calculated for the symmetric systems when the two polymer brushes faced each other. In addition, the authors fitted force-distance curves using the Hertz model (Eq. (18)) to determine the apparent Young's modulus of the polymer brushes [151]. According to the Alexander-de Gennes model, the forces between two approaching surfaces that are covered with polymer brushes depend on the grafting density of the polymer brushes and their thicknesses [3]. The thermal motion of a polymer chain at a surface is limited by the approach of another surface, resulting in a decrease in the entropy of individual polymer chains. Therefore, confining dangling polymer chains generates a repulsive force of entropic origin known as steric hindrance. In addition, the concentration of the polymer segments in the interfacial gap increases, leading to increased osmotic pressure [3]. The steric repulsion force between a spherical body and a planar surface that are both covered with an impenetrable brush of uniform grafting density, at distances smaller than twice the brush thickness, can be derived using Derjaguin's approximation to give [3,153]:

$$\frac{F(D)}{R} = \frac{16\pi}{35} k_B T L_0 \Gamma^{3/2} \left[7 \left(\frac{D}{2L_0} \right)^{-5/4} + 5 \left(\frac{D}{2L_0} \right)^{7/4} - 12 \right] \quad (21)$$

where R is the radius of the colloidal probe, L_0 is the thickness of the polymer brush on each side, Γ is the number of bound polymer chains per unit area (grafting density), and D is the separation between the two surfaces. This expression was modified by O'Shea et al. [154] to describe the interaction of a polymer brush with a bare surface (asymmetric systems) by replacing $2L_0$ in Eq. (21) with L_0 and halving the force to give [153,155]:

$$\frac{F(D)}{R} = \frac{4\pi}{35} k_B T L_0 \Gamma^{3/2} \left[7 \left(\frac{D}{L_0} \right)^{-5/4} + 5 \left(\frac{D}{L_0} \right)^{7/4} - 12 \right] \quad (22)$$

For $0.2 < D/L_0 < 0.9$, the above expression is roughly exponential and can be approximated by [156]:

$$F(D) = 50 k_B T R L_0 \Gamma^{3/2} e^{-2\pi D/L_0} \quad (23)$$

When applying this model to the data, it is assumed that the applied load fully compresses the polymer brush; therefore, data with applied forces higher than 3 or 5 nN have usually been considered [151,155]. Furthermore, the equations above were derived using a stepped profile in which the volume fraction of the undisturbed brush is assumed to be constant within the brush and zero elsewhere. In contrast, Milner, Witten, and Cates [157] used a more realistic parabolic profile for the volume fraction of the brush, as follows [158,159]:

$$\frac{F(D)}{R} = 4\pi P_0 \left[\frac{2L_0^*}{D} + \left(\frac{D}{2L_0^*} \right)^2 - \frac{1}{5} \left(\frac{D}{2L_0^*} \right)^5 - \frac{9}{5} \right] \quad (24)$$

with

$$P_0 = \frac{k_B T n}{2} \left(\frac{\pi^2 l^4}{12} \right)^{1/3} \Gamma^{5/3} \quad (25)$$

where n is the mean number of segments in the polymer chains, l is the length of a segment, and L_0^* is the equilibrium brush thickness. In this model, the thickness of the brush is slightly larger than that in the Alexander-de Gennes model: $L_0^* = (24/\pi^2)^{1/3} L_0 = 1.34 L_0$.

It should be noted that the mechanical and conformational properties of the polymer brush systems are functions of various factors, such as the ionic strength, pH, and temperature of the medium.

Polyelectrolyte polymer brushes generally shrink at high salt concentrations, according to the predictions of the scaling laws. In contrast, most zwitterionic polymer brushes, for example, exhibit an antipolyelectrolyte effect behavior where they don't shrink in high salt solutions, but their polymer chains swell alternatively when increasing the ionic strength of the solution [160]. Guo et al. used colloidal probe AFM (SiO_2 probe with a diameter of 5 μm) to estimate the grafting density and apparent Young's modulus (using the Hertz model) of two sulfobetaine brushes, methacrylamide and vinylbenzene prepared by SI-ATRP [155]. The results were presented in terms of the brush swelling responses to the ionic strength of the solution and temperature changes. The swelling response of the polymer brushes was determined by AFM imaging of the cross-sectional profiles which were created by gently scratching the substrates to remove the polymer brushes from them and was also determined by the in-situ liquid spectroscopic ellipsometry. Furthermore, measurements of the adhesion force between colloidal probes covalently modified with cyprid footprint proteins and the polymer brushes correlated well with the cyprid settlement data for the studied surfaces [155].

4.1.3. Nanotribological properties of abiotic surfaces

The tribological properties of contacting surfaces and interfaces have been characterized by measuring the friction or lateral forces using colloidal probe AFM. Such properties of abiotic surfaces and their wear resistance against rubbing can be of importance in biofoulant cleaning processes. In addition, many medical applications require the contact surfaces to be resistant to biofouling and wear at the same time. For example, Chang et al. studied the role of lubricin and hyaluronic acid, the molecular constituents of the synovial fluid, in the boundary lubrication of diarthrodial joints [161]. The friction force of these two molecules were measured at the microscale between a hydrophilic (hydroxyl-terminated) and a hydrophobic (methyl-terminated) surfaces in a physiological buffer solution. Their results indicated that the addition of hyaluronic acid did not alter the frictional behavior between the model substrates demonstrating that no synergistic effect on friction behavior existed in a physiological mixture of lubricin and hyaluronic acid. In contrast, a synergistic effect between lubricin and chemically grafted or immobilized hyaluronic acid resulted in enhanced lubrication and antiadhesive properties [162,163]. Zhang et al. investigated the effects of nano-/micropatterning and a hydrophobic coating on the adhesion and friction forces between the surface and a SiO_2 AFM colloidal probe [164]. They prepared different surfaces patterned by spin-coating of suspensions with polystyrene microspheres of two diameters (0.44 and 2.00 μm) and further modified the surfaces with a self-assembled monolayer of 1-octadecanethiol (ODT). The textured surfaces had lower friction and adhesion forces than flat surfaces at all applied forces, which was attributed to the reduced actual contact area. In addition, the ODT coating further decreased the friction and adhesive forces owing to the lower surface energy of the modified surface and the reduction in the meniscus-mediated (capillary effect) adhesion and friction between the interacting interfaces [164]. Man et al. fabricated nanotextured patterns of different morphologies on a Ni—Co coating using direct energy density-controlled laser writing [165]. The nanotribological properties of textured surfaces were characterized by AFM with a colloid probe to generate friction force versus load plots. Their results indicated that the nanotextured surfaces adjusted the surface friction forces compared with the unprocessed Ni—Co coating surface. In addition, half-elliptical patterns showed better tribological properties than hemispherical patterns [165]. Morgese et al. grafted brushes comprising equimolar amounts of PEG and poly(2-methyl-2-oxazoline) (PMOXA) or poly(2-ethyl-2-oxazoline) (PEOXA) onto SiO_2 surfaces to improve their lubrication characteristics. The effect of brush composition on the nanotribological properties of the surfaces was investigated via lateral force microscopy (LFM) using a silica colloidal probe [166]. Using variable angle spectroscopic ellipsometry and a quartz crystal microbalance with dissipation (QCM-D), the authors found an increase

in hydration and viscoelasticity, decreased friction, and improved lubrication characteristics for mixed brushes compared to their single-component counterparts (Fig. 7) [166].

4.1.4. Microscale wettability of abiotic surfaces

Wettability is a fundamental characteristic of abiotic surfaces that plays an essential role in their biofouling [167,168]. In this respect, the colloidal probe technique is one of the available methods that offers many ways to measure surface wettability. Babel et al. employed hydrophobic colloidal probes to characterize the wettability of mineral surfaces (chalcopyrite, pyrite, and quartz) at the microscale, showing that the overall order of the median values of $-F_{\text{max}}/R_{\text{particle}}$ distributions corresponds to the order of the macroscopic contact angles (Fig. 8) [169]. It was possible to differentiate between quartz and both chalcopyrite and pyrite which reflected the force distribution trend. The authors found that it is more challenging to distinguish between chalcopyrite and pyrite because of their similar contact angles. On the other hand, the use of the colloidal probe technique to measure particle-bubble interactions and extract the contact angle/wettability of the particle surface from the force–distance curves is well-documented in the literature [170].

4.2. Isoelectric points of trace amounts of protein biofoulants

The isoelectric point is the pH at which a particular molecule carries a net-zero charge. It is usually difficult to obtain the isoelectric points of small quantities of proteins using conventional methods. Guo et al. proposed a method for measuring the isoelectric points of trace amounts of proteins using colloidal probe AFM [171]. The method measures the adhesion forces as a function of pH using colloidal probes functionalized with the protein of interest against positively and negatively charged substrates prepared via layer-by-layer deposition of polyelectrolytes. According to this method, the isoelectric point of a protein is defined as the pH at which a transition is observed in the adhesion force measurements. Using four proteins with known isoelectric points (BSA, myoglobin, fibrinogen, and ribonuclease A), the measured isoelectric points had an estimated accuracy of ± 0.25 . The authors also measured the isoelectric point of a footprint of temporary adhesive proteins secreted by the barnacle cyprid larvae of *Amphibalanus amphitrite* to be in the range of 9.6–9.7 (Fig. 9). Unlike other conventional methods, the authors underlined that this method can be used for trace amounts of proteins (approximately 470 protein molecules), is not limited by protein solubility, and can be used for unknown proteins without knowing their amino acid sequences. In addition, the protein of interest can be immobilized on the colloidal probe using small deposits at a solid surface without the need to prepare protein solutions. However, this method requires stable engineered reference substrates that are either positively or negatively charged under a broad range of pH, and that maintain similar surface properties, such as composition, roughness, and hydrophilicity when the pH is varied. Furthermore, the interactions in the system should be dominated by electrostatic forces with relatively low non-Coulombic interactions in the background. Another concern may be the effect of covalent immobilization of the proteins at the surface of the colloidal probe and the consequent consumption of their functional groups on the protein's isoelectric point. However, the high accuracy of the isoelectric points estimated for the investigated proteins dispels such concerns and confirms the reliability of this method.

4.3. Phenotypic heterogeneity in the attachment behavior of bacteria at the population and single-cell levels

The development of single-cell level approaches, such as the colloidal probe technique (particularly cell probes, see Section 5.1), has enabled studies on the behavior of bacteria to understand how a single cell behaves in comparison to its population. Phenotypic fluctuations between bacterial cells have been attributed mainly to differential gene

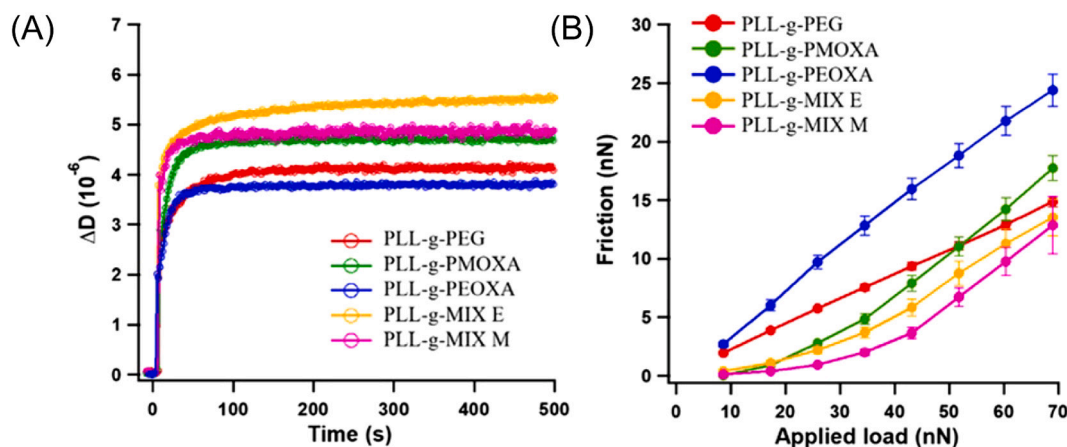


Fig. 7. (A) Change in QCM-D energy dissipation (ΔD) recorded during graft copolymer adsorption on SiO_x -coated sensors. (B) Friction force vs applied load profiles recorded by LFM of different brush films. Adapted with permission from [166].

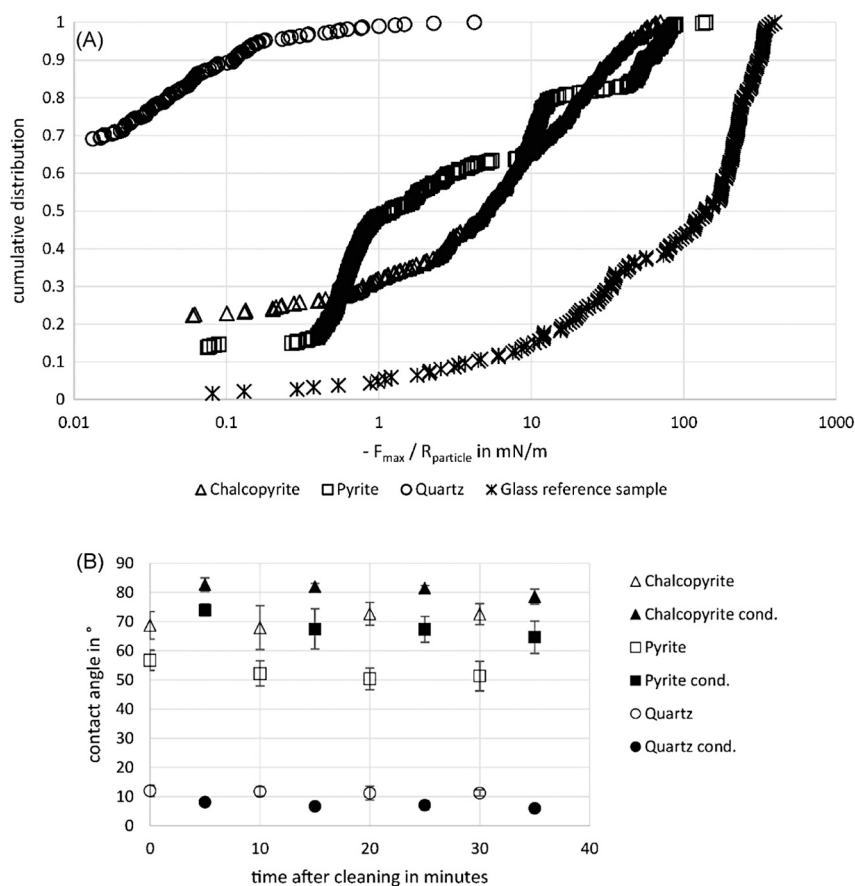


Fig. 8. (A) Cumulative force distributions of mineral surfaces, and (B) their time-dependent contact angles with and without potassium octylxanthate conditioning. The error bars represent the standard deviations of the contact angles. The comparison of adhesive force distributions shows a shift to higher adhesive forces with an increase in the macroscopic contact angle. Adapted with permission from [169].

expression and allelic alterations [172]. El-Kirat-Chatel et al. used bacterial colloidal probes to investigate correlations between bacterial attachment to surfaces at the population and cellular levels [172]. Three strains that varied in their phenotypic traits (TC5, TC10, and TC11) were studied to evaluate their adhesion to a hybrid block copolymer comprising a self-polishing coating and a fouling release coating (SPC-FRC) and an SPC block alone. The attachment assays of these three strains and the hybrid coatings were performed using a microplate assay combined with AFM and confocal scanning laser microscopy. The TC11

strain adhered to both surfaces the most at the population level, while TC5 was the most adhesive strain on both surfaces at the single-cell level. Their study revealed that adhesion forces at the cellular level do not always correlate with the adhesion behavior at the population level, because some bacteria are susceptible to phenotypic heterogeneity in their populations [172]. Therefore, it is recommended to apply appropriate statistical measures to the single-cell adhesion data by collecting force-distance curves using many independent cells ($n > 10$), including sets of independently prepared cells and target samples [173].

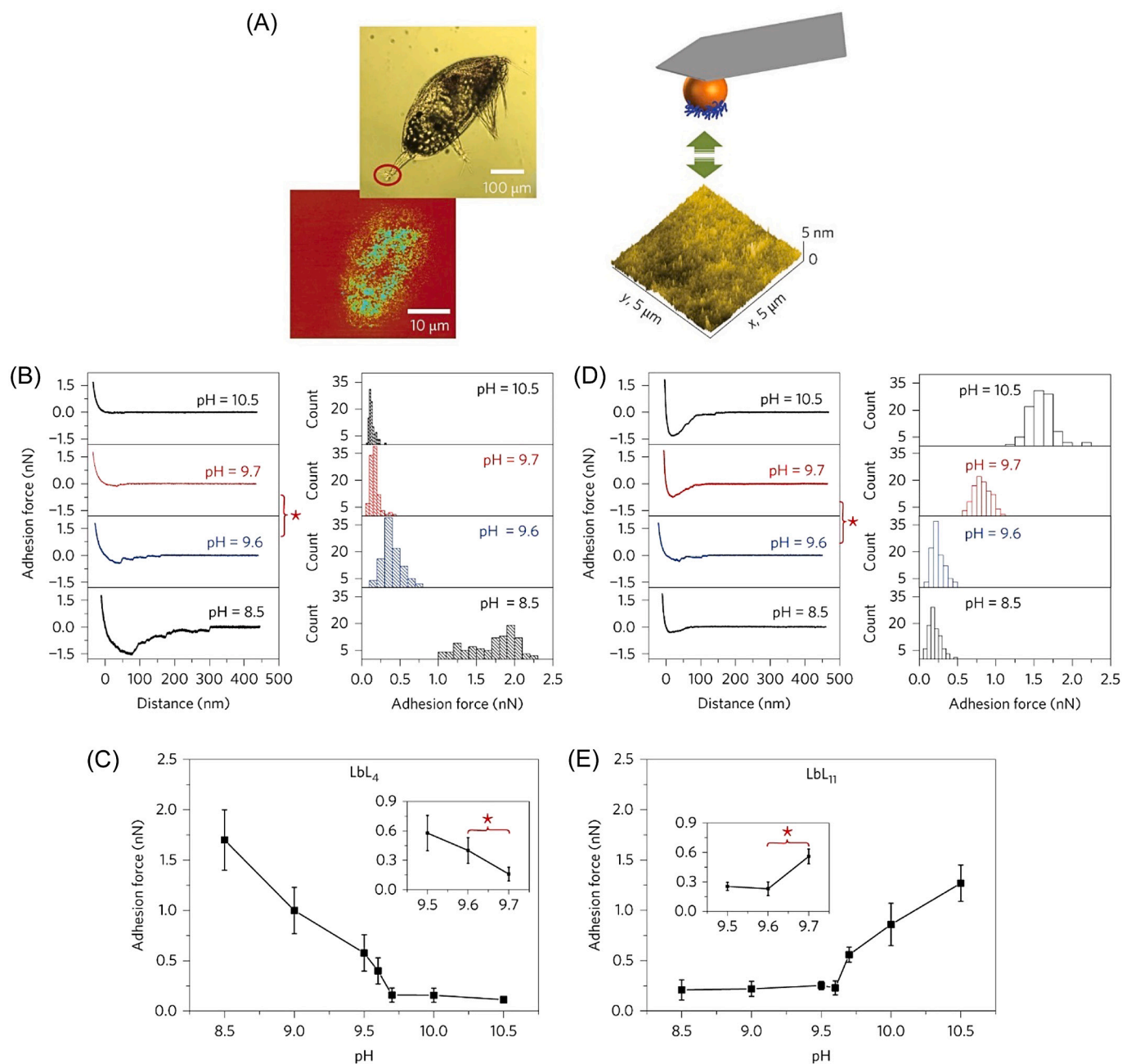


Fig. 9. Estimating the isoelectric point of barnacle cyprid footprint proteins using the functionalized colloidal probe technique. (A) Modification of a colloidal probe with barnacle cyprid footprint proteins. The area with the footprint protein deposits is labeled with a red circle. (B) Retraction curves of the footprint-protein-modified probe with the negatively charged surface LbL₄, (comprising six layers of poly(ethylene imine) (PEI) and seven layers of poly(isobutylene-*alt*-maleic anhydride) (P1)) and the corresponding histograms of the adhesion forces under various pH conditions. (C) Adhesion forces of the LbL₄ surface in the pH range 8.5–10.5. A significant adhesion-force change occurred in the range 9.6–9.7 (inset). (D) Retraction curves of footprint-protein-modified probes with the positively charged surface LbL₁₁, (comprising seven layers of PEI and seven layers of P1) under various pH conditions, and the corresponding histograms of the adhesion forces. (E) Adhesion forces of the LbL₁₁ surface in the pH range 8.5–10.5. A significant adhesion change occurred in the range 9.6–9.7 (inset), as marked with red stars, which indicates the pH value of the footprint protein. Adapted with permission from [171]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.4. Electrochemical force spectroscopy

Force and electrochemical measurements have recently been performed by simultaneously recording force–distance curves and Faraday-current–distance curves using electrochemically controlled conductive colloidal probes allowing quantitative electrochemical adhesion measurements. Such measurements can be of interest to study the biofouling of electrified interfaces such as electrodes. Different approaches have been developed to fabricate electrically insulated AFM cantilevers with

gold-coated colloidal spheres located at the end of the cantilever. Electrochemical polymerization has been used to coat these spheres to produce conductive polymer-coated colloidal probes and perform potential-dependent AFM force measurements [174]. A more robust boron-doped diamond colloidal electrode with an extended potential window and superior mechanical stability has also been developed [175]. These probes can be used for conductive AFM imaging of hard materials, such as diamonds, without deterioration of their conductive coating, which usually occurs with sharp commercial tips [175].

Furthermore, electrochemically tunable polydopamine (PDA)-modified probes with redox-switchable surface groups have been developed for adhesion measurements with various surfaces including treated gold surfaces and bacterial cells under potential control (Fig. 10) [176]. In addition, these PDA-modified probes prepared using the electrochemical pulsed deposition method ensured the formation of a uniform coating with nanoscale roughness on the probe surface for more precise force measurements. Although most conductive colloid fabrication approaches require sophisticated instrumentation such as focused ion beam milling to remove the insulation layer at the end of the cantilever, simpler approaches using polystyrene beads as templates have also been developed [177].

4.5. Interactions between abiotic surfaces and intermediate adhesive layers

The colloidal probe technique can be used to study the forces between abiotic surfaces and developed coating layers. Yang et al. investigated the effect of vanadium ions (V^{3+}) on the self-assembly of an intermediate adhesive layer composed of tannic acid (TA) and polyethylenimine (PEI) [178]. They used a PEI-coated silica colloidal probe on PEI deposited on a silica substrate in TA and TA/ V^{3+} solutions (Fig. 11). A significant adhesion force ($F/R \approx 0.65$ mN/m) was measured between the two PEI-coated surfaces in the presence of 10 mmol TA, which was attributed to a Schiff-base/Michael addition reaction between TA and PEI. In comparison, this force decreased to $F/R \approx 0.35$ mN/m in the presence of TA/ V^{3+} because the occupation of the catechol groups of TA by V^{3+} ions via a coordination reaction inhibits the Schiff-base/Michael addition reaction between TA and PEI. In addition, the colloidal probe technique was used to directly measure the adhesion force between the TA/PEI/V intermediate adhesive layer on a silica probe and a cellulose nanocrystal antifouling coating. The authors detected a strong adhesion force ($F/R \approx 0.62$ mN/m) owing to the synergistic effects of interactions such as electrostatic interactions, hydrogen bonding, and coordination reactions. Thus, the colloidal probe

can help optimize the assembly process of coatings and understand the involved reaction mechanisms.

5. Modification of colloidal AFM probes for force measurements of abiotic surfaces

The relatively large surface area of colloidal probes facilitates surface functionalization with materials or molecules of choice and their subsequent use in colloidal chemical force microscopy. Colloidal probes have been modified in various ways to simulate the behavior of biofoulants and their interactions with surfaces. In addition, modified colloidal probes have been used as indicators of the biofouling propensity of abiotic surfaces. The following sections describe modification procedures developed by the research groups.

5.1. Cell-immobilized colloidal probes (cell probes)

Cell (or biocolloid) probes have been employed to perform single-cell force spectroscopy (SCFS) measurements that allow quantification of the interaction forces at the single-cell level between a cell and a substrate of interest. The first SCFS experiments were performed by Bowen et al., in which a yeast cell (*Saccharomyces cerevisiae*) was attached to a tipless cantilever using cyanoacrylate or poly(lysine) glues where they indicated that SCFS is a relatively fast method for assessing biofouling potential [122,179]. Various immobilization methods have subsequently been developed to attach the cells of interest to AFM cantilevers. Examples include coating the cantilever with positively charged proteins, such as PEI [180] and poly-L-lysine [181], or using naturally derived proteins that bind strongly to various surfaces, such as PDA [172,182–185] and polyphenolic protein (Cell-Tak) [186]; hydrophobic coatings such as thiol monolayers [187]; chemical fixation such as glutaraldehyde treatment [180,188]; and covalent functionalization of cantilevers with lectin [189] or silanes [190]. Fluidic force microscopy (FluidFM) has recently emerged as a powerful technique for reversibly immobilizing single cells [191]. FluidFM combines AFM with

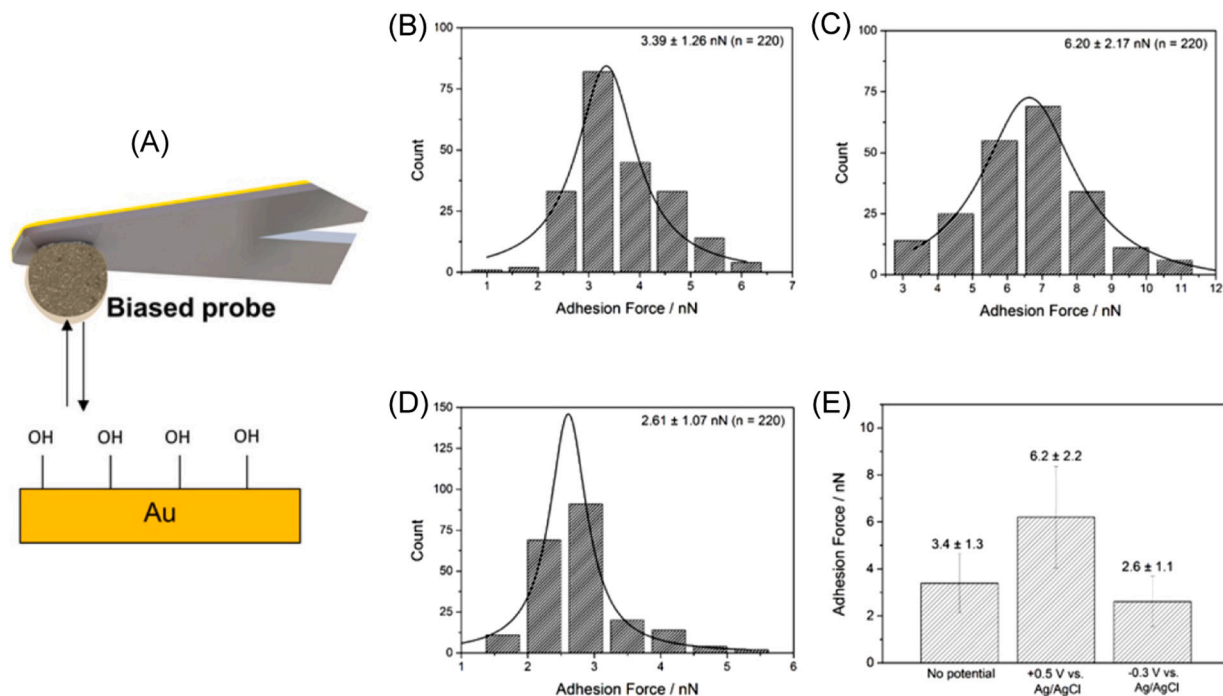


Fig. 10. (A) Schematic of electrochemical force spectroscopy of a plasma-treated gold surface. Adhesion force histograms measured using a PDA-modified colloidal AFM-scanning electrochemical microscopy probe with (B) no potential applied, and the probe biased at (C) 0.5 V and (D) -0.3 V versus Ag/AgCl in 0.1 M KCl. (E) Adhesion force versus the different applied potentials. Adapted with permission from [176]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

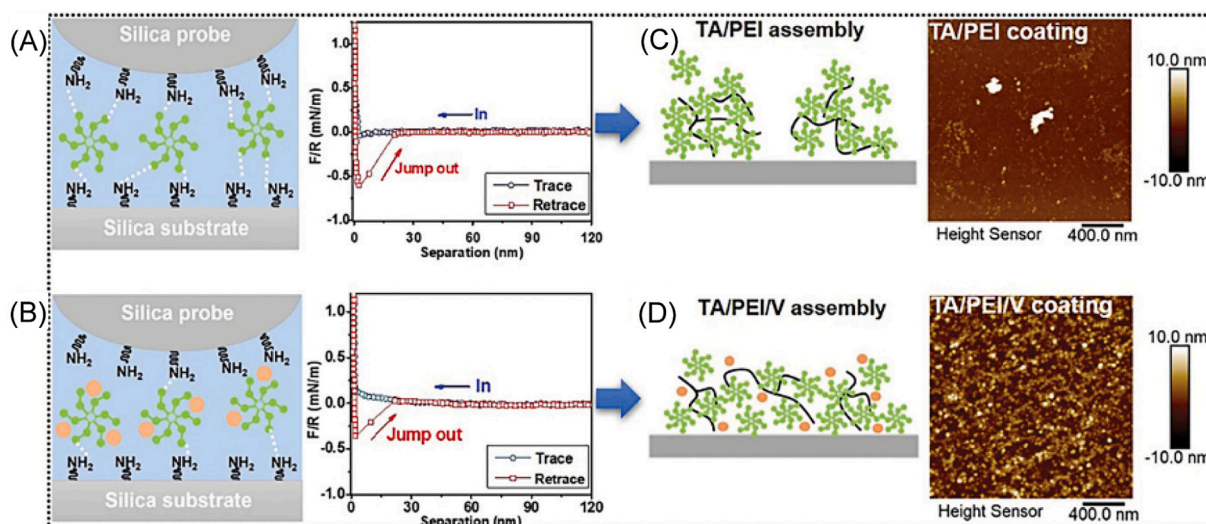


Fig. 11. Force–distance curves collected between a PEI-coated silica probe and a PEI-coated silica substrate in the presence of (A) 10 mmol TA and (B) 10 mmol TA and 10 mmol V^{3+} . Topographic AFM images of (C) TA/PEI coating and (D) TA/PEI/V coating. Reprinted (adapted) with permission from [178].

microfluidics and allows in situ single-cell immobilization by applying suction through a microfluidic cantilever connected to a pressure controller [192–194]. However, it requires more advanced instrumentation than conventional AFMs. FluidFM technology has also been used to prepare particle colloidal probes by in-situ aspiration of particles to the aperture of a hollow micro-channel AFM cantilever. This enabled overcoming of the size limitation of conventional colloid probe AFM where only particles with diameters of several micrometers (determined by optical resolution) could be used as probe particles. With FluidFM, the lower limit of the colloidal probe technique is extended by one order of magnitude, allowing particles with diameters well below 500 nm and/or particles with soft outer shells to be used as they have many major industrial applications [195,196].

Bacteria-coated probes have been prepared by immobilizing a confluent layer of bacterial cells from suspensions onto poly-L-lysine- or PEI-modified tipless cantilevers [181,197]. While a single-cell probe estimates the overall surface adhesion of an individual cell, multicellular-coated probes average the inhomogeneities observed from individual bacterial cells. However, the varying contact surface area and roughness of multicellular-coated probes limit their reproducibility.

Multiple studies have quantified the adhesion interactions between cell probes functionalized with unicellular microorganisms and various surfaces. Thwala et al. measured the interaction forces between *Pseudomonas putida* and *Bacillus subtilis* with different polymeric membranes and compared them to the extended XDLVO theory predictions and the steric interactions involved [184]. The more hydrophilic and negatively charged bacteria, *B. subtilis*, exhibited lower adhesion compared to *P. putida*, highlighting the contribution of hydrophobic and acid–base interactions to the membrane's bacterial adhesion. BinAhmed et al. investigated the interactions between *Pseudomonas fluorescens* and polysulfone UF membranes to gain insight into the molecular mechanisms underpinning membrane biofouling [185]. They showed that hydrophilic PDA coatings on the membrane reduced the bacterial adhesion for short contact times. However, these hydrophilic coatings were less effective in preventing adhesion during prolonged contact periods. The authors attributed this to bacterial surface macromolecules, which experience substrate-induced unfolding upon contact with the membrane. Arce et al. functionalized AFM cantilevers with live diatom cells (*Navicula species I*) and measured their adhesion to a hydrophobic elastomer, Intersleek, and a hydrophilic mica [198]. They showed that live diatom probes can be used to rapidly assess the anti-biofouling potential of different surfaces. Rodriguez-Emmenegger et al. measured the adhesion of *Yersinia pseudotuberculosis*, a common bacterial

pathogen, on seven types of antifouling polymer brushes grafted on glass via SI-ATRP [199]. They found that the maximum force required to separate the bacterial probe from the brushes decreased by 99% and acted over shorter distances compared to non-coated glass. In addition, the work required to detach the bacterium from the brushes was only 2 to 20% of that on glass. Shiimoto et al. investigated the adhesion of the tentacle that the cypris larva uses to explore suitable sites for settling [200]. The authors measured the adhesion force of the tentacle of a live cypris immobilized on a tipless cantilever with polymer brush-coated glass using scanning probe microscopy. Hydrophilic polymer brushes were prepared on cover glasses and silicon wafers via SI-ATRP of 3-(*N*-2-methacryloyloxyethyl-*N,N*-dimethyl) ammonatopropanesulfonate (MAPS) or 2-hydroxyethyl methacrylate (HEMA). After the cypris larva probe touched the surface for a few seconds, the cantilever was twisted according to the adhesion strength, and the lateral deflection of the cantilever was recorded (Fig. 12). The adhesion force was calculated by multiplying the lateral deflection of the cantilever by the torsional spring constant and the cantilever sensitivity. The adhesion behavior of the tentacles of live cypris on unmodified hydrophobic cover glass coated with a propylsilane monolayer gradually increased with time, reaching 30–60 μN after 14–20 days. In comparison, the hydrophilic poly(MAPS) and poly(HEMA) brush grafted surfaces showed very low adhesion over three weeks. The studies mentioned above demonstrate the potential of cell probes to predict the biofouling behavior of abiotic or modified abiotic surfaces.

5.2. Membrane material-modified colloidal probes

Colloidal probes fabricated using polymeric membrane materials have been effectively used to quantify the adhesion interactions between membranes and foulants. Moreover, membrane-coated colloidal probes can be used to predict membrane biofouling and modify filtration membranes. Several methods have been proposed prepare colloidal probes using polymeric membrane materials. For example, using a micromanipulator to immobilize a PVDF microsphere onto a tipless cantilever, followed by thermal sintering of the microsphere at the softening point of PVDF (i.e., 112–154 $^{\circ}\text{C}$), resulted in the formation of a strong bond between the cantilever and the microsphere [123,201–203]. Meng et al. prepared a membrane-coated colloidal sphere via the phase inversion method by dipping the colloidal silica probe into a casting solution of PVDF/PVA (PVA) or PVDF/PVP (PVP), followed by immersion in deionized water to replace the solvent [204,205]. Song et al. followed the procedure developed by Wang et al.

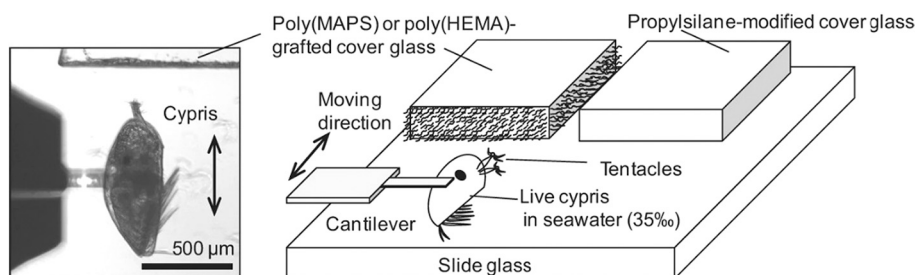


Fig. 12. Adhesion force measurement of a live cypris cell probe in seawater on cover glass surfaces modified with polymer brushes and propylsilane monolayer. Reprinted with permission from [200].

[123] to sinter polyamide and polyethersulfone (PES) microspheres onto AFM cantilevers [37].

5.3. Carboxylate-modified latex colloidal probes

Many researchers have used carboxylate-modified latex (CML) colloidal probes to simulate the behavior of different organic biofoulants and represent them in bulk solutions. Li et al. used a CML colloidal probe as a surrogate for HA, the major organic foulant found in natural waters [206]. Commercially available CML microspheres contain a high density of carboxylate groups and thus can simulate the predominant carboxylic acid functional groups of HA. The authors combined CML colloidal force measurements with NF membrane water-flux measurements to elucidate the mechanisms of organic fouling and chemical cleaning. Since it is challenging to use fouled membranes in AFM fluid cells without disturbing the fouling layer, the authors injected Suwannee River HA solutions into the fluid cell and left it to equilibrate for 30 min so that a fresh layer of foulant molecules could form on the clean membrane. Costa et al. used a CML colloidal probe as a surrogate for NOM to study the mechanisms of UF membrane colloidal NOM biofouling [207]. Lee et al. used CML to represent two model foulants, alginate and Suwannee River NOM (SRNOM), and investigated their interactions with fouled RO membranes [208]. They found a positive correlation between the foulant–foulant adhesion forces and the fouling rates of the biofoulants. Ang et al. used CML colloidal probes as surrogates for organic foulants (alginate and SRNOM) and measured the interfacial force between foulant in the bulk solution and foulant in the fouling layer on RO membranes [209]. In their study, a remarkable correlation was observed between the cleaning efficiencies and the foulant–foulant adhesion forces measured in the presence of the corresponding cleaning solutions. BinAhmed et al. used temperature-controlled CML colloidal probe force measurements to investigate the effect of feed-water temperature on the organic fouling of RO membranes [210]. On the other hand, Herzberg et al. used a CML colloidal probe as a surrogate for bacterial EPS, predominantly comprising carboxylic acid groups, to quantify the adhesion forces relevant to the membrane fouling [211]. The authors observed attractive jump-in forces during the approach of the CML probe in the presence of calcium ions, which the authors ascribed to complexation between calcium ions and EPS carboxylate groups. Moreover, the addition of calcium ions to the feed solution significantly increased the adhesion forces between the CML probe and EPS, indicating the critical role of EPS-calcium ion interactions in RO membrane biofouling.

CML colloidal probe force measurements have also been used to evaluate the enhanced biofouling resistance of modified water filtration membranes. For instance, Asatekin et al. used a CML colloidal probe to represent model organic foulants (BSA, alginate, and HA) because they contain many carboxylic acid groups [212]. In their study, a novel TFC NF membrane was prepared by coating a commercial PVDF UF membrane with an amphiphilic graft copolymer, poly(vinylidene fluoride)-*graft*-poly(oxyethylene) methacrylate (PVDF-*g*-POEM). CML colloidal probe measurements indicated the presence of steric repulsive forces, which likely prevented the adhesion of foulants to the TFC NF

membrane. In addition, the pull-off curves showed the absence of foulant adhesion to the TFC NF membrane. Lu et al. fabricated a TFC forward osmosis (FO) membrane by the in situ grafting of Jeffamine on a nascent polyamide active layer [213]. A CML colloidal probe was used as a surrogate for alginate macromolecules to confirm the biofouling resistance of the TFC FO membrane.

Other studies used CML as a surrogate for bacterial cells such as *E. coli* to test the anti-biofouling performance of modified membranes [214,215]. Castrillón et al. used a CML colloidal probe as a surrogate for alginate and HA and tested the improved biofouling resistance of a TFC FO membrane functionalized with PEG [216]. Ye et al. used a CML colloidal probe as a model for organic foulants and evaluated the biofouling resistance of their newly developed membranes [217]. Similarly, Perreault et al. used a CML colloidal probe as a model for biofouling and verified the anti-adhesive properties of their graphene oxide (GO)-functionalized TFC membrane [218]. Han et al. used a CML colloidal probe as a model for organic matter and measured its adhesion forces with hydroxyl- or epoxide-dominated GO membrane surfaces that were prepared by tuning the surface functional groups of GO membranes using a trace amount of -OH [219]. Kim et al. investigated the chemical heterogeneity of negatively charged polyamide RO membranes using positively charged aliphatic amine latex and CML colloidal probes [220]. Their findings suggested that membranes with a higher surface chemical heterogeneity exhibited a greater tendency toward biofouling.

5.4. Hydroxyl-modified colloidal probes

Yamamura et al. investigated the affinity of carbohydrate-like materials for microfiltration (MF) membranes using hydroxyl-modified polystyrene colloidal probes [221]. In their study, polybead-hydroxylate microspheres were used as surrogates for carbohydrate-like substances because of the hydroxyl-rich nature of carbohydrate materials, and polybead-carboxylate microspheres were used to represent humic substances. Stronger adhesion forces with the MF membranes were measured with the hydroxyl-modified colloidal probes than with the carboxyl-modified colloidal probes. The authors attributed this to strong hydrogen bonds in the case of the hydroxyl-modified microspheres, because the higher pKa value of the hydroxyl groups (9–13) compared with that of the carboxyl groups (3–6) prevented the dissociation of the hydroxyl groups at the measurement pH of 6.8. In addition, the affinity of the hydroxyl-modified microspheres was higher for the PVDF membrane than for the PE membrane, which was attributed to the high electronegativity of PVDF. Similarly, Zhang et al. used hydroxyl colloidal spheres as a surrogate of alginate to investigate the mechanism of fouling on a functionalized multiwalled carbon nanotube mixed-matrix FO membrane under various inorganic components with a high ionic strength [222].

5.5. Model foulant-functionalized colloidal probes

Physical or chemical functionalization of colloidal probes with a model foulant of interest has been performed to improve the

understanding of the molecular mechanisms of fouling. Colloidal probes have been modified by simple coating with a foulant of interest or covalent chemical functionalization which can be completed before or after attaching colloidal microspheres to the cantilever. In addition, the use of these modified colloidal probes makes it possible to measure the actual interaction forces between the foulant and the surface and between the foulant molecules themselves.

Comparative studies have been conducted to understand the complex phenomena of biofouling and investigate the affinity of surfaces toward various biofouling-causing macromolecules. Most studies have focused on the biofouling of filtration membranes. For example, sodium alginate has been shown to cause more severe biofouling of UF membranes than HAs. However, the adsorption of sodium alginate has been found to be largely reversible, as evidenced by backwashing tests [223]. Biofouling has been reported as the key contributor to the fouling of RO wastewater reclamation membranes, and proteins, humic substances, and polysaccharides as the main components of organic fouling [224]. Biopolymers, which are composed of hydrophilic organic matter, including carbohydrates and proteins, have been reported to be the major constituents that cause irreversible fouling in UF and MF membranes. In addition, the hydrophilic fractions of organic matter have been related to irreversible fouling, mainly because of their larger sizes (>100 kDa) and gelling properties [225,226]. In contrast, hydrophobic fractions show a higher propensity for fouling evolution in membrane systems because of their smaller size and adsorption in internal pores [223,227]. The following sections categorize modification strategies according to the type of foulant used.

5.5.1. Polysaccharide-functionalized colloidal probes

The accumulation of dissolved ions, organic molecules (e.g., polysaccharides and proteins), and other organic compounds in the marine environment contributes to the formation of conditioning films on solid surfaces. These conditioning films can significantly alter the physicochemical properties of substratum surfaces prior to bacterial adhesion, including surface tension, hydrophobicity, charge, and roughness, and can either enhance or inhibit initial bacterial adhesion [228–230]. Moreover, polysaccharides and polysaccharide-like matter are known to play pivotal roles in the biofouling of surfaces such as membranes [231]. Polysaccharide-like substances are neutrally charged and interact with surfaces either through hydrogen bonding or in colloidal form to create a cake/gel layer on the surface [232]. Additionally, the polysaccharide component of the microbial biofilm matrix provides many advantages to the cells in the biofilm, including adhesion, protection, and structure [233]. The majority of the biofilm matrix is composed of a viscous mixture of extracellular polysaccharides [234]. For example, the matrix of *Pseudomonas aeruginosa* is mainly composed of polysaccharide synthesis locus (Psl), pellicle (Pel), and alginate polysaccharides [234]. In contrast, proteins such as CdrA (an extracellular adhesin) act as struts between matrix components, adding strength to the structural scaffold, while the extracellular DNA in the matrix supplies further structural rigidity [234]. A higher alginate content in the EPS of *Pseudomonas aeruginosa* was shown to cause an increase in both the rate and extent of the attachment of EPS [235]. In addition, polysaccharides are a major component of the soluble microbial products (SMP) present in wastewater effluents that cause biofouling of surfaces such as membranes [211,236]. Therefore, sodium alginate has been widely investigated as a model polysaccharide for biofouling research.

Different modification methods have been exploited to functionalize colloidal probes with polysaccharides in order to investigate the affinity between the polysaccharides and filtration membranes. Wang et al. investigated typical organic foulants, including alginate, which were adsorbed onto a PVDF microsphere probe, sintered onto a cantilever to prepare foulant-coated colloidal probes [123]. Miao et al. used the same procedure to prepare alginate-coated colloidal probes and investigated the effect of pre-ozonation on the fouling behavior of UF membranes [237]. Tiraferri et al. prepared an alginate-coated colloidal probe by

immersing a CML colloidal probe in an alginate solution [238]. The foulant-coated colloidal probe was used to measure the interfacial adhesion forces of a hydrophilic silica nanoparticle-coated TFC FO membrane. Wang et al. coated colloidal probes with sodium alginate (rich in $-\text{COOH}$) and chitosan (rich in $-\text{NH}_2$) and measured their adhesion forces on PVDF and GO-modified PVDF membranes under various pH conditions [239]. Similarly, Zhu et al. prepared alginate-coated colloidal probes according to the procedures described by Wang [123] and Miao [240] and investigated the combined fouling behavior of FO membranes with alginate and TiO_2 nanoparticles in Ca^{+2} solutions [241]. The alginate and FO membrane adhesion forces were characterized using alginate-coated colloidal probe, while a TiO_2 -coated colloidal probe was used to measure the adhesion force between TiO_2 and the alginate-fouled membrane.

5.5.2. Protein-functionalized colloidal probes

Proteins are another major component of the microbial biofilm matrix and also contribute to the conditioning of solid surfaces. Proteins and protein-like materials have been identified as problematic foulants that cause fouling of surfaces such as membranes [232]. Protein-like foulants are amphoteric materials that interact, for example, with membrane surfaces through dipole interactions or in colloidal form, resulting in cake/gel layer formation [232]. BSA is a well-studied protein and is often selected as a model protein in biofouling studies of surfaces. For instance, Bowen et al. coated colloidal silica probes by immersing a silica microsphere attached to a cantilever in BSA solution for 3 h at 25 °C, which resulted in a single layer of firmly bound protein on the silica surface [122]. Han et al. followed a similar procedure by immersing a colloidal probe in a 1 g/L BSA solution for 24 h at 4 °C [242]. Hashino et al. used commercially available BSA-modified silica colloidal probes (Corefront, Japan) to study the biofouling of poly(ethyleneco-vinyl alcohol), PES, and PVDF membranes [243]. Miao et al. glued a BSA-modified microsphere (Micromod Partikeltechnologie GmbH Co., Germany) to a tipless cantilever and used it to investigate the effect of hydration forces on the protein fouling of membranes at various pH values [244]. Ma et al. also glued a commercial BSA-modified microsphere to a tipless cantilever and used it to explore the combined inorganic-organic (SiO_2 -BSA) fouling mechanisms of UF membranes [245]. Tiraferri et al. coated CML colloidal probes with BSA and other model foulants by soaking them in 2 g/L organic foulant solutions for at least 24 h at 4 °C to prevent organic degradation. Wang et al. functionalized silica microspheres coated with 2 nm of Cr and 50 nm of Au with self-assembled BSA monolayers as analogs for protein foulants that have both hydrophilic and hydrophobic units with positive and negative charges [246]. These colloidal probes were used to evaluate the antifouling performance of poly(3,4-ethylenedioxythiophene) and its zwitterionic derivative, synthesized via oxidative chemical vapor deposition.

Wang et al. investigated the effect of pH on the BSA fouling behavior of different hydrophilic PVDF UF membranes (i.e., PVDF/PVA and PVDF/PVP) [247]. The pH of the solution affected the surface charges and particle sizes of the BSA foulants, as well as the surface charges of the membranes. In addition, the solution pH significantly impacted the polar component of the surface tension (γ^{AB}). In contrast, it had an insignificant impact on the Lifshitz-van der Waals component of the surface tension (γ^{LW}). The rate of flux decline was the lowest under alkaline conditions for both membranes. However, under acidic conditions, the rate of flux decline increased but was always lower for PVDF/PVA than for PVDF/PVP. In addition, the adhesion force measurements by BSA-coated colloidal probes with BSA-fouled membranes were similar for the PVDF/PVA and PVDF/PVP membranes. At the same pH, the membrane-BSA adhesion force measured by the membrane-coated colloidal probe was lower for PVDF/PVA than for PVDF/PVP because of its more hydrophilic surface. Their XDLVO analysis demonstrated that the polar or Lewis acid-base interactions play a dominant role in membrane biofouling.

BSA has also been covalently immobilized on AFM colloidal probes. Xu et al. chemically immobilized BSA onto a sharp AFM tip as follows [248]. The AFM cantilever was first treated with oxygen plasma and then immersed in a 10 mM 3-aminopropyltriethoxysilane toluene solution for 2 h at room temperature. The amine-functionalized tip was then placed in glutaraldehyde solution (50% in water) for 30 min, followed by a reaction with BSA in phosphate buffer (PBS) at a pH of 7.4 for 40 min. The BSA-functionalized AFM tip was used to investigate the enhanced anti-biofouling of an organosilane-GO-functionalized PVDF hybrid UF membrane. Similarly, Zhu et al. covalently immobilized BSA and lysozyme proteins onto AFM colloidal probes as follows [249]. After plasma cleaning, colloidal probes were vapor-deposited with 3-aminopropyltrimethoxysilane (APTMS). The introduced amine functional groups were then reacted with glutaraldehyde, which served as a crosslinking agent to immobilize the protein. Finally, the treated colloidal probes were immersed in protein solutions and rinsed with PBS.

Zhang et al. used chemical force spectroscopy in conjunction with colloidal probes to quantify the evolution of the adhesion force of proteins on hydrated surfaces in water. They used BSA-functionalized colloidal probes to study the adhesion forces on surfaces with continuously varying hydrophobicity/hydration layers under varying pH, ionic strength, and HA concentration [250]. Hydrophobic interactions were the dominant interactions between BSA and the model substrates, and the passivation caused by hydration layers had a counteracting effect on the hydrophobic interactions. Furthermore, the authors tested hydrophobic and hydrophilic modified colloidal probes on model surfaces of varying hydrophobicity. The adhesion force for the hydrophobic probe was higher than that of the hydrophilic one due to the weak passivation from the hydration layer, which increased with increasing the hydrophobicity of the substrate. In comparison, the adhesion force was extremely weak for the hydrophilic probe. It did not change with the hydrophobicity of the substrate, mainly because of the passivation of the hydration layer. However, the change in the hydration layer with the change in the hydrophilicity of substrates was not demonstrated in their study.

Miao et al. investigated the mitigation mechanism of ozonation in the fouling of UF membranes by casein, which is abundant in dairy wastewater [251]. PVDF microspheres were sintered onto AFM cantilevers, and casein was adsorbed onto the microspheres to prepare casein-coated colloidal probes.

Guo et al. investigated the adhesion of adhesive proteins secreted by barnacle cyprid larvae (a marine fouling organism) of *Amphibalanus amphitrite* to model surfaces with variable hydrophilicity [252]. Their findings indicated stronger adhesion forces on a hydrophobic surface (21 ± 2 nN) compared to those of a hydrophilic surface (7.2 ± 1 nN). The low adhesion observed for hydrophilic surfaces pointed to the hydrophobic character of the protein and suggested that hydrogen bonding is not a controlling factor for protein interactions with surfaces. However, no significant disparities were found in the fractional cyprid settlement, indicating that surface wettability alone is not a key parameter to prevent settlement (Fig. 13) and that other factors, such as the surface charge, should be considered to interpret the cyprid settlement behavior.

5.5.3. Extracellular polymeric substances-functionalized colloidal probes

Surface-attached microorganisms live in self-produced three-dimensional biofilms of hydrated EPS mainly composed of biopolymers, such as polysaccharides, proteins, nucleic acids, and lipids. EPS mediates adhesion of cells to surfaces and their cohesion in biofilms [28]. Whether in bound or soluble form, EPS are significant foulants for submerged surfaces and membrane systems. Bound EPS consists of proteins, polysaccharides, nucleic acids, lipids, and HAs, which are found at or outside the cell surface [253]. In comparison, soluble EPS or SMP are organic compounds that are released into the solution from substrate metabolism (substrate-utilization-associated products) and

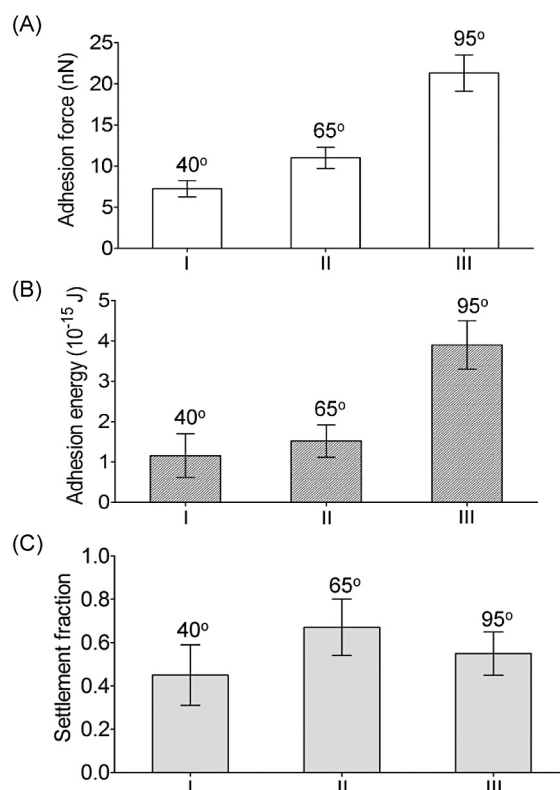


Fig. 13. (A) Adhesion forces, (B) adhesion energy, and (C) settlement assay of barnacle cyprid secretion on different model surfaces with variable hydrophilicity indicated by the contact angle of a water droplet. Reprinted (adapted) with permission from [252].

biomass decay (biomass-associated products) [253]. For instance, bound EPS has been reported to be a major sludge floc component and major foulant in membrane bioreactor systems. SMP is a major fraction of EfOM associated with NOM [254]. Both SMP and NOM play important roles in membrane fouling and flux decline in wastewater treatment [254]. SMP-based fouling usually causes pore-blocking owing to their soluble state, whereas EPS-based fouling results in the formation of a cake layer owing to their colloidal condition, generating a biofilm [255]. Polysaccharide-like components of EPS and SMP have been reported to contribute more to biofouling than protein-like materials [211,256]. To elucidate the interaction mechanisms between EPS and different model surfaces, EPS-modified colloidal probes have been used to perform AFM force measurements. For example, Ou et al. coated a colloidal probe with EPS using a layer-by-layer procedure, first proposed by Aubry et al. that uses a positively charged coating as an intermediate layer between the probe and the organic of interest [257,258]. They used iron oxide prepared by adding small amounts of NaOH to 10 mM FeCl₃ until the pH of the solution reached 7. The iron oxide-coated probe was then immersed in an EPS solution (50 mg/L) for 24 h and rinsed with deionized water [257]. Yu et al. prepared algae-related foulant-modified AFM colloidal probes (M-EPS-modified probes) [259]. The M-EPS solution was obtained by centrifuging an algal sludge mixture and collecting the supernatant. The AFM Cr—Au probe was then soaked in the M-EPS solution overnight to establish adsorption equilibrium and dried with N₂ gas. Zaouri et al. coated colloidal silica probes with high molecular weight compounds of *Skeletonema costatum* (a diatom responsible for marine algal blooms and irreversible membrane fouling) extracellular organic matter using a layer-by-layer procedure with iron oxide as an intermediate layer [125]. Furthermore, glutaraldehyde has been used to covalently incorporate cyprid footprint proteins on colloidal silica probes functionalized with APTMS [252,260].

5.5.4. Natural organic matter/humic acid-functionalized colloidal probes

NOM in natural waters is a complex mixture of colloidal particulates and dissolved organic materials generated by the chemical and biological degradation of plant and animal residues [261,262]. NOM has a wide range of molecular weights and functional groups (e.g., phenolic, hydroxyl, carbonyl, and carboxylic acid groups), and is usually divided into hydrophilic, hydrophobic, and transphilic fractions [263]. The hydrophobic fraction represents approximately 50% of dissolved organic carbon (DOC) with a larger molecular weight. The hydrophilic fraction, which is non-humic, contains 25–40% of DOC with a smaller molecular weight (e.g., polysaccharides, amino acids, proteins). Finally, the transphilic fraction constitutes approximately 25% of DOC, with molecular weight between that of the hydrophobic and hydrophilic fractions. A significant fraction of NOM originates from humic substances and accounts for 60–70% of DOC in soils and 60–90% of DOC in most natural waters [263]. Humic substances are recalcitrant anionic molecules with low-to-moderate molecular weight. They include aromatic and aliphatic components with predominantly carboxylic acid and phenolic functional groups [264]. Humic substances generally consist of HA, fulvic acids (FA), and humin, all of which can cause severe fouling that is sometimes irreversible compared to other constituents of NOM [261]. EfOM, on the other hand, can be defined as the non-settleable organic components of secondary-treated wastewater [265]. An investigation of humic substances in films deposited on solid surfaces from natural waters using electron impact, chemical ionization, and secondary ion time-of-flight mass spectrometry revealed a high degree of similarity with the spectra of purchased HAs (Fluka and Sigma), as well as with reference HA and FA (International Humic Substances Society), demonstrating that humic substances are the dominant foulants in natural water [266].

To gain a better understanding of the fouling mechanisms of abiotic surfaces by NOM, force measurements were carried out using colloidal probes modified with different NOM fractions and from different sources. For example, CML colloidal probes have been coated by immersing them in an organic foulant solution (e.g., 2000 mg/L of SRNOM for at least 24 h at 4 °C) to study the antifouling performance of membranes [231,267]. Similarly, a PVDF microsphere sintered to an AFM cantilever was coated by adsorbing EfOM fractions onto its surface [123,201,268]. Gutierrez et al. modified a silica colloidal probe in a layer-by-layer procedure using iron oxide as the positively charged intermediate layer [269]. Silica probes were immersed in an iron oxide solution for 12 h, rinsed with water, and then immersed for 12 h in a 1 mM NaHCO₃ solution containing Gartempe River or Brittany River NOM at concentrations of approximately 100 mg C/L. Finally, the NOM-coated colloidal probe was rinsed with deionized water and immediately used for AFM measurements [269]. Qu et al. immersed a colloidal probe in a concentrated NOM solution for 24 h [270]. The colloidal probe was stored at 4 °C for one week to strengthen the connection between the cantilever and glued NOM microsphere [270].

Several studies have also investigated interfacial interactions using HA-coated colloidal probes. Some of these studies sintered a PVDF microsphere onto a cantilever and then coated HA onto its surface [123,271,272]. Other researchers have covalently functionalized colloidal probes with HA to investigate the interactions between HA foulants and membrane surfaces. Johnson et al. used a N-(3-dimethylaminopropyl)-N'-ethylcarbodiimidehydrochloride (EDC) crosslinking agent to couple amine groups on amine-modified silica microspheres with carboxylic acid groups of HA and employed it to quantify adhesion forces with polymer membranes modified with a polymeric bicontinuous microemulsion coating in high-purity water and model textile dye wastewater [273].

Meng et al. investigated the effect of solution pH and the hydrophilicity of a PVDF UF membrane on the adhesion forces between membrane-HA and HA-HA using membrane-coated and HA-coated colloidal probes, respectively [204]. The membrane was prepared by mixing two kinds of hydrophilic polymers, polyvinylalcohol (PVA) or

polyvinylpyrrolidone (PVP) (PA is more hydrophilic than PP), with the PVDF polymer. Linear correlations were found between the adhesion force, normalized flux decline rate, and the irreversible fouling index of the membranes. The adhesion forces were greater in acidic environments than in alkaline or neutral environments and weaker for a more hydrophilic membrane. Their results showed that the solution pH and hydrophilicity of the membrane had a significant effect on the measured contact angles and particularly on the polar component of surface tension (γ^{AB}). An XDLVO analysis agreed with the trends of the colloidal probe measurements and revealed that the membrane-HA interactions were dominated by polar or Lewis acid-base interactions, which could be effectively reduced by improving the hydrophilicity of the PVDF UF membrane or reducing the solution acidity.

6. Biofouling mitigation and the AFM colloidal probe investigations of the anti-biofouling performance of modified abiotic surfaces

As discussed in Section 3, biofouling is controlled by a complex interplay among the primary interaction forces between the biofoulants/cells and abiotic surfaces submerged in aquatic systems, namely, the EL, LW, and AB forces. Consequently, biofouling mitigation strategies through modification of abiotic surfaces have to target these interplaying primary forces and weaken or reverse them (make them repulsive). The EL forces are usually ignored as they diminish at high ionic strengths or with pH changes that neutralize the surface charges of biofoulants and surfaces. In addition, strategies based on increasing the surface potential of abiotic surfaces would specifically repel only similarly charged biofoulants. However, oppositely charged biofoulants will be attracted to the surface. Thus, these strategies will be restricted to systems that contain either positively or negatively charged biofoulants. In addition, the EL interactions create repulsive barriers on the way while the biofoulants/cells migrate toward the surface, which sometimes can be merely overcome by the hydrodynamic flow forces in the medium or by the motility of microorganisms. The EL forces at contact distances are usually insignificant compared to LW and AB interactions.

The LW forces, on the other hand, depend on the combined Hamaker constant (A_{132}) of the biofoulant (1), abiotic surface (2), and aqueous medium (3) approximated by [107]:

$$A_{132} = \left(\sqrt{A_{11}} - \sqrt{A_{33}} \right) \left(\sqrt{A_{22}} - \sqrt{A_{33}} \right) \quad (26)$$

where A_{ii} is the individual Hamaker constants of materials 1, 2, and 3, given by [112]:

$$A_{ii} = 24\pi l_0^2 \gamma_i^{LW} \quad (27)$$

where l_0 is the minimum equilibrium distance between two parallel flat layers of material i , usually taken as 0.157 nm [112]. The Hamaker constant represents the non-geometrical contribution to the force of interactions, which depends solely on the properties of the molecules (e.g., polarizability and characteristic frequency of molecules). For dissimilar materials 1 and 2 immersed in liquid medium 3, a negative Hamaker constant can be attained and thus a repulsive LW interactions arise between materials 1 and 2 immersed in liquid medium 3, when [274,275]:

$$A_{11} < A_{33} < A_{22} \text{ or } A_{11} > A_{33} > A_{22} \quad (28)$$

or when

$$\gamma_1^{LW} < \gamma_3^{LW} < \gamma_2^{LW} \text{ or } \gamma_1^{LW} > \gamma_3^{LW} > \gamma_2^{LW} \quad (29)$$

The Hamaker approach which derives the Hamaker constant between condensed bodies from their molecular properties has some limitations. For example, it ignores the screening effect and the non-additivity of the molecular forces. This has led to the development of the Lifshitz macroscopic approach. In the Lifshitz approach, the LW

forces between continuous bodies are derived based on the bulk properties such as the dielectric constants (also called relative permittivities) and refractive indices (refractive index is the square root of the dielectric constant). If the Lifshitz approach is used to calculate the Hamaker constant instead of the thermodynamic approach, which relies on the apolar LW component of the surface tension [107], A_{132} will be given by:

$$A_{132} \approx \frac{3}{4} kT \left(\frac{\epsilon_1 - \epsilon_3}{\epsilon_1 + \epsilon_3} \right) \left(\frac{\epsilon_2 - \epsilon_3}{\epsilon_2 + \epsilon_3} \right) + \frac{3h}{4\pi} \int_{\nu_1}^{\infty} \left(\frac{\epsilon_1(i\nu) - \epsilon_3(i\nu)}{\epsilon_1(i\nu) + \epsilon_3(i\nu)} \right) \left(\frac{\epsilon_2(i\nu) - \epsilon_3(i\nu)}{\epsilon_2(i\nu) + \epsilon_3(i\nu)} \right) d\nu \quad (30)$$

where ϵ_i is the static dielectric constant of material i , $\epsilon\epsilon(i\nu)$ are the values of ϵ at imaginary frequencies ($i\nu$), and $\nu_n = (2\pi kT/h)n = 4 \times 10^{13} \text{ ns}^{-1}$ at 300 K. The first term in Eq. (30) gives the zero-frequency of the LW interactions and includes the Keesom and Debye dipolar contributions, while the second term gives the dispersion energy and includes the London energy contribution. In this case, A_{132} will be negative and thus the resulting LW interactions will be repulsive, when [107]:

$$\epsilon_1 < \epsilon_3 < \epsilon_2 \text{ or } \epsilon_1 > \epsilon_3 > \epsilon_2 \quad (31)$$

As biofouling of abiotic surfaces usually occurs in aqueous systems, according to Eqs. (29) and (31), to get repulsive LW forces and prevent biofouling, water is supposed to have an apolar surface tension and a dielectric constant that fall between those of the two-interacting materials (i.e., the biofoulant and abiotic surface). However, water has a γ^{LW} of 21.8 mJ/m^2 , while most organic and inorganic solids and/or materials have a γ^{LW} value in the range between 35 and 40 mJ/m^2 [114,115]. Moreover, the dielectric constant is a frequency-dependent parameter, but the quoted dielectric constant values in the literature tend to be attained at low frequencies. However, it is the values across the whole frequency range that affect the interactions, particularly the ones at high (UV) frequencies. At these high frequencies, the refractive index and hence the dielectric constant of water is very low [107]. It is thus difficult to find a material with permittivity values lower than that of water that can repel the biofoulants. There are a number of reports on nanomaterials developed for optical coatings with quoted refractive indices lower than that of water [276,277]. However, this appears to be because they are porous materials that have an average refractive index value between the material and air and therefore they would give a very different value in water. According to the discussion above, repulsive LW interactions between biofoulants and abiotic surfaces immersed in water seem hard to be attained. This is mainly for applications when the changes in the aqueous medium to modify the surface tension, such as by adding surfactants, increasing ionic strength, or adding organic solvents, are not possible [275]. However, the minimum LW interactions are expected to be attained when the abiotic surface is modified in a way to adjust its properties and make them close to those of water ($A_{22} \approx A_{33}$).

In Eq. (17), and according to the standard sign convention, a positive sign for the interaction energies between biofoulants and abiotic surfaces immersed in water indicates repulsion, while a negative sign indicates attraction. In most cases, it is not possible for the first term in Eq. (17) ($\Delta G_{132}^{\text{LW}}, 0$) to be positive (repulsive), as discussed above. If the second term $\Delta G_{132}^{\text{AB}}, 0$ dominated over the first one and was repulsive (positive), then $\Delta G_{132}^{\text{IF}}$ can be positive, and biofouling can be unfavored or suppressed (assuming negligible EI forces). However, the pronounced asymmetry of the polar properties of the most existing hydrophilic materials in favor of their electron donicity compared with the polar properties of water results in a strong AB repulsion [112]. Van Oss credited this to the prevalence of oxygen in the Earth's atmosphere and surface crust, making most of the common materials either oxides or hydroxides, which are predominantly electron-donors [115]. The polar repulsion between monopolar molecules or materials immersed in water can override (i.e., $\Delta G_{132}^{\text{IF}} > 0$) the attractive hydrophobic interactions due to the hydrogen bonding cohesion energy of water (102 mJ/m^2 at 20°C) and the van der Waals interactions between water molecules and between the monopolar materials, when the electron donor parameter (γ^-)

of the monopolar materials 1 and 2 is larger than 28.3 mJ/m^2 (assuming that γ^{LW} for both biofoulants (1) and abiotic surfaces (2) $\approx 40 \text{ mJ/m}^2$, and ignoring the EI interactions) [112].

Numerous efforts have been dedicated to modifying the properties of abiotic surfaces and enhancing their biofouling resistance (for reviews, see [278–281]). These modification pathways are usually based on the addition of materials with low affinity toward biofoulants, materials that possess antimicrobial/bactericidal activities, or both [282–284]. However, there is a general consensus that hydrophilicity makes surfaces less prone to biofouling because of the formation of a hydration layer that provides an energetic barrier to the adsorption of foulants [212,285–287]. Therefore, many attempts have focused on increasing the hydrophilicity of surfaces by incorporating hydrophilic materials, such as nanoparticles (NPs), within the substrate matrix [288] or by modification of the surface as a post-fabrication process, including surface coating and surface grafting [289]. Zhao et al. have pointed out the misperception that confuses hydrophilicity with the contact angle of water in the literature and demonstrated that the water contact angle on a membrane surface does not correlate with its fouling, but the hydrophilicity of the surface quantified by the following equation does [290]:

$$\Delta G_{\text{iwi}} = -2\gamma_{\text{iw}} = -2(\gamma_{\text{iw}}^{\text{LW}} + \gamma_{\text{iw}}^{\text{AB}}) = -2\left(\sqrt{\gamma_{\text{i}}^{\text{LW}}} - \sqrt{\gamma_{\text{w}}^{\text{LW}}}\right)^2 - 4\left(\sqrt{\gamma_{\text{i}}^+ \gamma_{\text{i}}^-} + \sqrt{\gamma_{\text{w}}^+ \gamma_{\text{w}}^-} - \sqrt{\gamma_{\text{i}}^+ \gamma_{\text{w}}^-} - \sqrt{\gamma_{\text{i}}^- \gamma_{\text{w}}^+}\right) \quad (32)$$

where i is for the surface, and w is for water. Furthermore, the authors also illustrated that the electron donor surface tension parameter (γ^-) of the membrane surface also correlated with the membrane adhesive fouling and could be used as an efficacious indicator of membrane biofouling. They also observed from sensitivity analyses of the total interaction energy ($\Delta G_{132}^{\text{TOT(XDLVO)}}$) between foulants and membrane surface in the short range of separation that the total energy became repulsive once γ^- of the membrane surface exceeded a certain value that they termed the critical membrane surface γ^- . This value was found under their study conditions to be 38.95 mJ/m^2 . This is consistent with van Oss et al., who mentioned that for the polar repulsion to prevail between monopolar molecules or materials immersed in water, it needs to overcome the attractive hydrophobic interactions due to the hydrogen bonding cohesion energy of water and the van der Waals interactions between water molecules and between the monopolar materials [112]. Their critical value was estimated as 28.3 mJ/m^2 according to their assumptions (as discussed earlier). Therefore, the biofouling of surfaces can be attenuated by enhancing γ^- to above its critical value. Moreover, the hydrophilicity of organic and inorganic solids strongly correlates with high γ^- values. The boundary between hydrophilicity and hydrophobicity occurs when the difference between the apolar attraction (LW) and the polar repulsion (AB) between molecules or surfaces of material i immersed in water is equal to the cohesive polar (AB) attraction between water molecules (Eq. (32)) [291]. When $\Delta G_{\text{iwi}} > 0$, the interaction of material i with water dominates and the material is hydrophilic, while when $\Delta G_{\text{iwi}} < 0$, the polar cohesive attraction between water molecules dominates and the material is hydrophobic. The binding energies between less hydrophilic molecules or surfaces and water are mainly apolar (LW), while the binding energies between the more hydrophilic molecules or surfaces and water (with increasing hydrophilicity) are of an increasingly mixed LW and AB nature [112]. In Eq. (32), ΔG_{iwi} will be positive (i.e., repulsion), if the second term: ($\gamma_{\text{iw}}^{\text{AB}} < 0$, and $|\gamma_{\text{iw}}^{\text{AB}}| > \gamma_{\text{iw}}^{\text{LW}}$ [112]. As most hydrophilic polar surfaces are monopolar, a higher γ^- will give rise to a higher binding energy with water and increase the hydrophilic repulsion between surfaces and biofoulants.

In general, surfaces that resist the adsorption of proteins have been described as hydrophilic, electrically neutral, and possessing hydrogen-bond acceptor groups and not hydrogen-bond donor groups [292].

These characteristics are in line with the hydration resistance theory of protein adsorption because they only pertain to the interactions between water molecules and functional substrate ligands [293]. However, several self-assembled ligands have been reported to exhibit properties

that partially contradict those mentioned above in this paragraph and yet are resistant to protein adsorption. This has been explained by extending the above general rules to account for steric repulsion and the entropic contribution to protein adsorption resistance. Thus, in addition

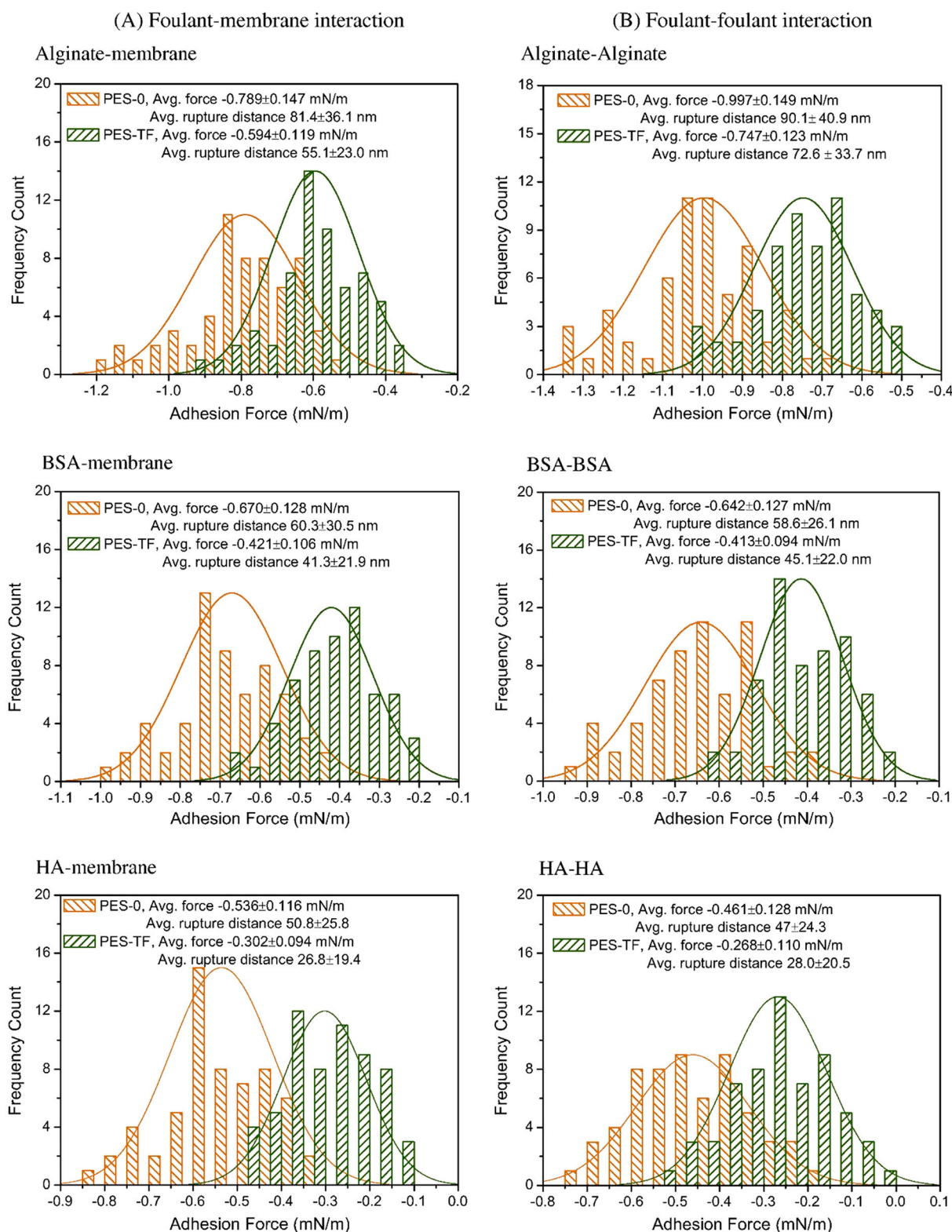


Fig. 14. Adhesion interactions of (A) foulant-membrane and (B) foulant-foulant interactions measured by AFM in contact mode. The different plots show the interactions between membrane surfaces (unmodified (PES-0) in yellow, modified (PES-TF) in green) and model biofoulants alginate, BSA, and HA. Reprinted with permission from [286]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to the above properties, the ligands on the substrate must be relatively large, conformationally flexible, and possess polarizable groups, which together account mainly for the graft density of ligands [293]. These rules agree with van Oss's interpretations above, where an electrically neutral condition is necessary to acquire negligible electrostatic interactions, and hydrogen-bond acceptor groups are one type of electron donor groups.

Bio-inspired micro-nano structured superhydrophobic surfaces have also gained interest because of their intrinsic antifouling and self-cleaning properties as a result of the combined effect of the low surface energy and the air layer entrapped in the solid-liquid micro-nano groove interfaces [294–296].

Many studies have focused particularly on the biofouling of filtration membranes because their operation requires them to be in direct contact with water systems in various configurations. In addition, the permeate drag force due to flow through the membrane tends to pull dissolved material toward the membrane feed water facing surface of the selective layer. An agreement has been observed between foulant–membrane AFM colloidal probe adhesion force measurements and membrane filtration performance in terms of flux decline and cleaning efficiency [123,204,267]. Moreover, correlations between the foulant–foulant colloidal probe adhesion forces and the fouling rate, as well as between the cleaning efficiencies and the colloidal probe adhesion forces measured in the presence of the corresponding cleaning solutions, have been reported [208,209]. This demonstrates the possibility of using the mean AFM adhesion forces of the model foulants measured by the colloidal probe technique to predict the biofouling propensity of a surface. Consequently, several studies have tested the antifouling performance of newly developed biofouling-resistant surfaces by measuring the AFM interfacial force using model foulant-modified colloidal probes. For instance, Hilal et al. used the photoinduced grafting technique to modify a PES MF membrane for biofouling mitigation [297]. The two hydrophilic monomers 2-acrylamido-2-methyl-1-propanesulfonic acid and quaternary 2-dimethylaminoethylmethacrylate were grafted onto the PES surface for their high hydrophilicity and bactericidal activity. The adhesion forces of the unmodified and modified membranes were measured using a silica colloidal probe to be 39.46 ± 6.13 mN/m and 24.75 ± 5.01 mN/m, respectively (with a loading force of 55.9 ± 5.7 nN). In addition, the number of bacterial cells (*E. coli*) able to proliferate from countable colonies was found to be much lower on the modified membrane than on the unmodified PES membrane. Li et al. investigated the antifouling mechanism of PES UF membranes impregnated with self-assembled TiO₂ nanoparticles owing to their superhydrophilicity, high photocatalytic ability, and antimicrobial properties [286]. CML colloidal probes modified with organic foulants (alginate, BSA, and HA) were used to estimate the foulant–membrane and foulant–foulant adhesion forces. The foulant–membrane adhesion forces provide information about the initial attachment of foulant molecules to the membrane surface, while the foulant–foulant adhesion forces indicate the adhesion strength of foulant molecules already adhered to the membrane surface. Weaker foulant–membrane and foulant–foulant adhesion forces with narrower distributions were observed for the composite membrane than for the pure membrane, indicating the efficacy of the modification in mitigating membrane biofouling (Fig. 14). Tiraferri et al. used colloidal probes coated with BSA and other model foulants to measure the interfacial interactions between foulants and a superhydrophilic TFC FO membrane with a silica nanoparticles-functionalized surface [267]. Weaker adhesion forces were measured for the superhydrophilic membrane than for the control TFC polyamide membrane. The antifouling properties of the superhydrophilic membrane were ascribed to the tightly bound hydration layer at the membrane surface, as well as the neutralization effect of the positively charged silica nanoparticles that covered the carboxyl groups of the polyamide membrane, preventing calcium ions from bridging with carboxyl-rich foulants.

Liang et al. prepared a PVDF UF membrane via post-fabrication

tethering of surface-tailored superhydrophilic silica nanoparticles onto a poly(methacrylic acid)-grafted PVDF membrane surface [298]. The foulant–membrane interaction forces were quantified using colloidal probes coated with sodium alginate, SRNOM, or BSA. The foulant–membrane interfacial forces of the functionalized membrane were weaker than those of the unmodified membrane. These results agreed with the fouling performance of the membranes, demonstrating the capability of AFM force measurements to predict membrane fouling behavior. Castrillón et al. synthesized TFC FO membranes comprising microporous polysulfone support over which a polyamide selective layer was fabricated by interfacial polymerization, and functionalized them with PEG [216]. PEG is known to confer biofouling resistance to surfaces via repulsive steric forces [299]. To graft PEG, ethylenediamine was reacted with the acyl chloride groups of the polyamide layer, forming a selective layer rich in primary amine groups. The resulting amine-rich layer was then grafted with PEG diglycidyl ether, whose epoxide groups readily react with primary amines. To assess biofouling propensity, the interaction forces between a CML colloidal probe used as a surrogate for alginate and the membrane surfaces were quantified. The adhesion forces on the PEG-grafted polyamide membrane exhibited a narrower distribution and were significantly weaker (-0.15 ± 0.12 mN/m) compared to the control polyamide membrane (-0.81 ± 0.54 mN/m). However, the rupture distances were statistically similar for both membranes (83.5 ± 58.8 nm and 78.8 ± 53.8 nm for the control and PEG-functionalized membranes, respectively) [216].

Another modification strategy was applied by Herzberg et al. via graft polymerization with oppositely charged zwitterionic monomers on a PVDF membrane, namely, negatively charged 3-sulphopropyl methacrylate and positively charged [2-(methacryloyloxy)ethyl]-trimethylammonium [300]. Adhesion force measurements were performed using CML colloidal probes as an analog for EPS in water and 100 mM NaCl. The adhesion forces between the CML colloidal probe and the modified membrane were significantly lower than the unmodified membrane, and the adhesion forces of both membranes were lower in NaCl than in pure water. The authors attributed the lower adhesion force of the modified membrane to its increased hydration (swelling). The weaker adhesion force measured in NaCl was explained by favorable electrostatic interactions between the CML particles and the NaCl solution at the expense of interactions between the CML particles and the membrane surface. In addition, the reduced biofouling propensity of the modified membrane was attributed to the increased hydrophilicity of the PVDF surface, and most likely owing to its neutral charge properties. Similarly, Schon et al. grafted zwitterionic PSBMA brushes onto silicon and glass substrates via SI-ATRP [145]. AFM tips were chemically functionalized with fibronectin as a model for the nonspecific surface adsorption of cyprinid temporary adhesive, and with RGD peptide motifs covalently attached to PMAA brushes as biomimics of cellular adhesion receptors. They highlighted the fundamental role of zwitterionic polymers in resisting biofouling, which was attributed to the hydration layer that forms at the interface due to the electrostatic forces between the zwitterions and bulk water, rather than hydrogen bonding, which is the case of PEG coatings. Their results showed a significantly reduced nonspecific adhesion for fibronectin-functionalized tips to PSBMA-modified substrates (0.25 ± 0.18 nN and 0.03 ± 0.003 nN for thin and the thick brushes, respectively) compared to the adhesion to bare gold (2.3 ± 0.75 nN) and self-assembled ODT monolayers (1.3 ± 0.75 nN). No significant adhesion was observed for PMAA-RGD-modified colloidal probes on the PSBMA-modified silicon substrate [145]. The anti-biofouling property of membrane surfaces modified with zwitterions has also been attributed to their significantly enhanced electron donor component (γ^-) of the surface tension indicating that they are likely to be ionized to donate electrons in aqueous solutions [301].

Liu et al. compared the biofouling resistance of two different types of highly hydrophilic materials (i.e., SiNPs and zwitterionic polymers) that are used for the surface modification of polyamide TFC membranes [287]. Dense layers of SiNPs and zwitterionic polymer brushes were

grafted onto the membrane surfaces via dip-coating with aminosilane-functionalized SiNPs (SiNP-TFC) and SI-ATRP of sulfobetaine methacrylate (PSBMA-TFC), respectively. Adhesion force measurements were performed using a CML colloidal probe as a model for organic foulant rich in carboxylic acid groups. A high proportion of non-adhesion events (48%) was observed in the PBS buffer for the pristine TFC membrane, with an average intermolecular force of -0.09 ± 0.21 mN/m. After modification with SiNPs, the mean adhesion force increased by approximately 36% (-0.14 ± 0.23 mN/m), while the PSBMA-TFC membrane exhibited weaker adhesion forces (-0.07 ± 0.14 mN/m). Although both modified membranes had the same degree of surface hydrophilicity enhancement, the stronger adhesion of the SiNP-TFC membrane was attributed to the electrostatic attraction between the negatively charged foulants and the positively charged SiNPs. On the other hand, the much lower adhesion of the PSBMA-TFC membrane suggests that the neutrally charged zwitterionic PSBMA brush layer created an energetic barrier for the adhesion of foulants, reducing the membrane fouling propensity. In comparison, the pristine membrane in a solution containing 50 mM NaCl and 0.5 mM CaCl_2 showed markedly stronger adhesion forces (-0.31 ± 0.20 mN/m) than those in the PBS solution, indicating that calcium ions aggravate fouling by complexation with carboxylic acid groups. However, the mean adhesion forces for both modified membranes showed an approximately 61% reduction compared with the pristine membrane. This was attributed to the coating layers on the modified membranes, which prevented complexation between the carboxylic acid groups on the membrane and the CML probe. This indicates the importance of considering the precise environment in which the membranes are likely to be used when performing the colloidal probe force measurements. In addition, the improved fouling resistance of the PSBMA-TFC membrane compared to that of the SiNP-TFC membrane in static adsorption tests with proteins and bacteria, as well as in dynamic FO fouling experiments, was rationalized by electrostatic interactions and the high density of surface carboxylic acid groups on SiNP-TFC that are still accessible to foulants. In contrast, the zwitterionic polymer brushes shielded the surface carboxylic acid groups and provided steric hindrance against foulant adsorption due to significant hydration.

Another membrane surface modification strategy is to use nano-imprinting to fabricate line-shaped surface patterns on membrane distillation (MD) membrane surfaces [302]. Xie et al. used a BSA-modified spherical silicon colloidal tip to measure the interaction forces between BSA and surface-patterned or pristine commercial polytetrafluoroethylene membranes. However, because the nanoimprint procedure does not alter the membrane surface chemistry, adhesion forces from a limited number of spots will not adequately detect changes in the adhesion characteristics. Therefore, adhesion maps were generated to determine the spatial distribution of the force interactions between the BSA-modified colloidal probe and the patterned MD membrane surface. The adhesion maps showed that BSA adhesion with the patterned membrane (-0.301 ± 0.13 μN) was lower than that with the pristine membrane (-0.495 ± 0.14 μN). Hydrophobic interactions were the dominant interactions between BSA and the MD membranes. During filtration, the BSA particle size in the feed solution gradually increased from 5.9 to 824.6 nm, reaching the size of the pattern feature. Therefore, this study demonstrated that highly ordered features on the MD membrane surface cause a decrease in the number of favorable interaction sites for BSA protein, thereby minimizing BSA deposition onto the membrane. Zhang et al. investigated the effect of the polymorphous structures of a PVDF membrane on its organic fouling behavior [303]. Two PVDF membranes with similar pore structures and distinct polymorphs were prepared and investigated using model foulants (HA, BSA, and DEX). The interaction forces between the foulant-coated particle probes and each membrane surface were measured using AFM with an initial loading force of 5 nN. The foulant-membrane interaction forces had a much wider distribution with large negative values for the α -PVDF membrane compared to that of the β -PVDF

membrane. The enhanced antifouling behavior of the β -PVDF phase was attributed to the stronger electron donor monopolarity ($\gamma^- = 25.3$ mJ/ m^2) compared to that of the α -PVDF phase ($\gamma^{\text{AB}-} = 3.3$ mJ/ m^2). Pan et al. developed a mussel-inspired coating (PDA-PAA-PMTAC) consisting of carboxyl and quaternary ammonium moieties with tunable surface charges that enable long-range electrostatic repulsion interactions with emulsion droplets of different charges [304]. The surface charge-tunable antibiofouling coating via changes in solution pH resulted in an outstanding and regenerable separation performance of both cetyltrimethyl ammonium bromide (CTAB) and sodium dodecyl sulfate (SDS)-stabilized emulsions at the desired pH conditions. At pH 4, the coating acquired a positive charge and could repel CTAB-stabilized emulsions, while at pH 10 it acquired a negative charge and could repel SDS-stabilized emulsions. The interaction forces between oil droplets (toluene) with CTAB or SDS adsorbed at the interface and PDA-PAA-PMTAC-coated silicon substrates were measured using the AFM droplet probe technique with the droplet attached to the AFM cantilever. Decreasing the solution pH resulted in a positively charged coating and strong electrostatic repulsion was observed between the CTAB-stabilized toluene droplets and the polymer coating. In contrast, increasing the solution pH resulted in a negatively charged coating, and strong electrostatic repulsion was observed between the SDS-stabilized toluene drops and the polymer coating surface. The force profiles obtained in the droplet probe measurements were fitted with a theoretical model based on Stokes-Reynold-Young-Laplace equations to describe the deformation of the oil drop surface and correlate it with hydrodynamic and disjoining pressures. More examples of the studies in the literature that assessed the antifouling performance of modified abiotic surfaces in comparison with their pristine counterparts are summarized in Table 1.

7. Current challenges and future perspectives of the biofouling potential/anti-biofouling performance investigated by the colloidal probe technique

Probe preparation is one of the challenges in the colloidal probe technique, especially those prepared using conventional methods with particles ≤ 5 μm . Improper preparation can result in glue-coated or contaminated particles, glue-altered particle properties, and agglomerate attachment of particles [316]. Adhesive-free fabrication methods for colloidal probes have also been developed using melting or sintering approaches [317,318]. However, these irreversible methods apply only to materials with melting points lower than that of silicon or silicon nitride cantilevers, such as borosilicate glass. In addition, the limited resolution of optical microscopy, which is frequently used in combination with micromanipulation systems for gluing particles, cannot detect the topography of the particle to check whether there is any defect or protrusion on the particle surface, and additional complementary techniques such as SEM might need to be applied. Another challenge can be posed by the roughness of the colloidal probe, which leads to uncertainties in the contact area between the probe and the surface, and thus affects the accuracy of the measured adhesion forces [319].

The lifetime of AFM cantilevers is usually limited by attaching colloidal particles using the conventional methods. In addition, possible contamination of these cantilevers with glue can pose a further challenge, despite some recycling methods for detaching the particle and cleaning the cantilevers have been proposed such as the plasma cleaning method [320]. Another possible challenge arises from the contamination of colloidal probes or deterioration of their functionalization layer after interaction with the sample surface. Cleaning or decontaminating colloidal probes can be more challenging compared with sharp tips, as traditional cleaning techniques such as plasma and UV-ozone can be destructive to the colloidal probe and may cause detachment of the particle or deterioration of its functionalized layer. Nonetheless, a non-destructive method using mechanical scrubbing with supersharp brushes for cleaning colloidal probes has been proposed [321]. Furthermore, loss of loosely attached particles during performing

Table 1

Summary of the studies that assessed the antifouling performance of modified abiotic surfaces using the colloidal probe technique.

Abiotic surface material	Modification strategy	Colloidal probe	Normalized colloidal AFM mean adhesion force for the pristine surface (mN/m)	Normalized colloidal AFM mean adhesion force for the modified surface (mN/m)	Maximum loading force used in the AFM colloidal probe force measurement (nN)	Biofouling/ cleaning studies	Ref
Polyethylene terephthalate films	Biomimetic nanoscale nipple arrays or moth-eye structures made from acrylic resin and about 100 nm in height (MOSMITE™)	SiO ₂	in water: 3.9 ± 0.33 in air: 105 ± 31	in water: 1.0 ± 0.26 in air: 13 ± 4.6	20	–	[305]
Silicon wafers	Biomimetic shark-skin surface (Sharklet AF™) with different surface chemistries, such as –(OH) and –(CH ₃) end groups	DOPA (a model adhesive protein)-functionalized SiO ₂	The peak force is in the range of for: Si: (1.2–7.7)	The peak force is in the range of for: –CH ₃ : 0.7–2.1 –OH: 0.4–1.0	3–4	Compared with QCM-D: Less attachment of protein to the OH-terminated Sharklet AF™ than CH ₃ -terminated, which is in accordance with the weaker force measurements under static conditions.	[306]
L80 carbon steel	Electroless nickel-phosphorus coating	Asphaltenes-coated SiO ₂	2.7 (in 1 mM NaCl)	0.25 (in 1 mM NaCl)	10	Bulk fouling tests in asphaltenes-coated silica suspension. The bulk fouling results agreed well with the surface force measurement results.	[307]
Electron-beam-deposited iron substrates	Electroless nickel-phosphorus coating	Toluene (a model oil) drop probe	16 (in 100 mM NaCl and with 10 ppm of asphaltenes)	4.1 (in 100 mM NaCl and with 10 ppm of asphaltenes)	–	Bulk fouling tests in oil-in-water emulsions with asphaltenes. Iron substrate was fouled in the toluene-in-water emulsion with asphaltenes, while the coated substrate remained clean.	[308]
Polyamide thin-film composite (TFC) FO membrane	Graphene Oxide Functionalization of membrane surface	Carboxylated latex particle	0.49 ± 0.47	0.15 ± 0.23	1	After 1 h, deposition and viability of <i>Pseudomonas aeruginosa</i> cells on GO-TFC were reduced by 36% and 30%, respectively. After 24 h, membrane biofouling was reduced by 50%.	[218]
Polished single-crystal silicon wafers	Hybrid hairy Janus particles with hydrophilic poly(poly(ethylene glycol) methyl ether methacrylate) (P(PEGMA)) and hydrophobic polydimethylsiloxane (PDMS) or poly-(monomethacryloxypropyl-terminated polydimethylsiloxane) (P(PDMSMA)) polymers grafted on the opposite sides of the particles	SiO ₂	–	Spherical particles pure-coated with: -P(PEGMA): 0 -P(PDMSMA): 8 Platelet like particles pure-coated with: -P(PEGMA): 0 -P(PDMSMA): 23 Spherical particles coated with PDMS and P (PDMSMA): 52%/48% distribution of force-distance curves. Platelet like particles coated with P(PEGMA) and P(PDMSMA): 44%/56% distribution of force-distance curves.	20	Chemoluminescence assay that quantifies the adhering viable marine biofilm-forming bacteria <i>Cobetia marina</i> . The Janus particle coatings significantly reduced bacterial retention under both static and dynamic conditions independent of the particle geometry.	[309]
Whatman tetrafluoroethylene (PTFE) membrane filter	Liquid-infused (LI) slippery surface	Polydopamine (PDA)-coated silica	1.89 ± 0.19 ($F_{adh}/R \sim 0.378$), (in 1 mM NaCl solution)	Strong jump-out due to the capillary force with oil interface/the force sensitivity limit is exceeded/ jump-out distance:	5.6 51.5	LI-PTFE prevented PDA deposition, as evidenced from the unchanged morphology, wettability, and slippery characteristics before and after PDA deposition. Gentle water rinse washed off the adsorbed PDA particles on	[310]

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Table 1 (continued)

Abiotic surface material	Modification strategy	Colloidal probe	Normalized colloidal AFM mean adhesion force for the pristine surface (mN/m)	Normalized colloidal AFM mean adhesion force for the modified surface (mN/m)	Maximum loading force used in the AFM colloidal probe force measurement (nN)	Biofouling/ cleaning studies	Ref
				~1.92 μm (at 5.6 nN loading force), ~2.10 μm (at 51.5 nN) (in 1 mM NaCl solution)		LI-PTFE due to the ultralow friction as verified by lateral force microscopy (LFM).	
Polyamide UF membrane	Grafting thermoresponsive poly(N-isopropylacrylamide) (PNIPAM) chains on a bromine-containing polyamide layer.	BSA-coated silica	0.92 $\pm$ 0.33	0.62 $\pm$ 0.29	–	Static adsorption and dynamic fouling experiments with BSA revealed that PNIPAM chains improved the deposition and adhesion resistance. In addition, PNIPAM chains shrank when the temperature was above the lower critical solution temperature (LCST), forming a buffer layer between the membrane and pollutants enhancing the cleaning efficiency.	[311]
Polystyrene, glass and stainless steel	Block copolymer coating of polystyrene and a zwitterionic peptide comprising of lysine and glutamic acid.	SiO ₂	6.3 (W=F/2 π R = 1 mJ/m ²)	~0	–	The brush-coating reduced EPS adsorption onto polystyrene surface in 100 mM NaCl by ~70% (mass), reduced the rate of bacterial attachment by >80%, and inhibited the development of thick biofilms. However, QCM-D results demonstrated that the EPS molecules penetrate between the chains of the brush and adsorb onto the polystyrene surface.	[312]
Silicon wafers	ABA triblock copolymer with A as a catechol-anchor and B as a hydrophilic zwitterionic block (poly(dopamine methacrylamide)-poly[(sulfobetaine methacrylate)-co-(2-(dimethylamino)ethyl methacrylate)]-poly(dopamine methacrylamide) (PDMA-P(SBMA-co-DMAEMA)-PDMA))	Polystyrene colloidal probe	–	~0 (a stronger repulsive force against compression/enhanced steric stabilization of the surface)	–	HRP-conjugated anti-IgG adsorption tests demonstrated that triblock polymer-coated surface displayed excellent protein reduction performance compared with a bare silicon surface. In addition, bare silicon exposed to BSA-FITC emitted a high intensity of fluorescence (LCSM) and a much smaller amount of protein was detected on the triblock copolymer-coated silicon wafer.	[313]
Silicon wafer, stainless steel mesh and non-woven fabric	Bio-inspired zwitterionic copolymer: 2-methacryloyloxyethyl phosphocholine (MPC) and dopamine methacrylamide (DMA) copolymer (PDMA-co-MPC) grafted coating.	Polystyrene colloidal probe	–	~0 (a repulsive force with the probe due to highly swollen nature and hydrophilicity of PDMP copolymers)	–	The amount of protein (peroxidase-labeled goat antibody human IgG antibody) adsorption was significantly reduced for the modified surfaces/ A high fluorescence intensity (LCSM) on the silicon wafer. While the fluorescence intensities on modified surfaces were significantly reduced/Oil repelling enhancement.	[314]
Polythiourethane (PTU)	Polythiourethane/ZnO (5 wt% t-ZnO) composite material	<i>Staphylococcus aureus</i> bacteria (~1 μm of diameter), cells immobilized on silica colloidal probes. <i>Candida glabrata</i> yeast cells (~4–5 μm in diameter)	<i>S. aureus</i> : 0.237 $\pm$ 0.289 <i>C. glabrata</i> : 4.881 $\pm$ 1.744	<i>S. aureus</i> : 0.086 $\pm$ 0.12 <i>C. glabrata</i> : 2.684 $\pm$ 1.066	1	–	[315]

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Table 1 (continued)

Abiotic surface material	Modification strategy	Colloidal probe	Normalized colloidal adhesion force for the pristine surface (mN/m)	Normalized colloidal AFM mean adhesion force for the modified surface (mN/m)	Maximum loading force used in the AFM colloidal probe force measurement (nN)	Biofouling/cleaning studies	Ref
immobilized on tipless cantilevers.							

measurements due to frictional forces with the sample can be another challenge that necessitates duplicating the experiment. Additionally, calibrating the spring constant of large and soft colloidal probes requires additional consideration and attention. The thermal tune method may underestimate the spring constant for large colloidal probes. Thus, a correction factor needs to be measured and multiplied by the spring constant. Alternatively, the spring constant of the tipless cantilever can be first measured and then corrected for the loading conditions of the particle. The deflection sensitivity can then be calculated from the estimated spring constant [78]. When calibrating soft colloidal probes, the deflection sensitivity should not be measured using the conventional contact approach.

With the ripening of the revolutionary fields of microfabrication and 3D printing, more advanced and accurate technologies in atomic force microscopy are likely to evolve in the coming years with practical solutions to the challenges mentioned above. Nevertheless, the FluidFM, developed in 2009, is a robust technology that is continuously growing to accommodate a broader range of applications.

8. Concluding remarks

Biofouling is a complex process that involves multiple interactions of different types and ranges. Therefore, a fundamental understanding of the nature of interactions between biofoulants/microbial cells and abiotic surfaces is essential for developing effective antifouling strategies. Colloidal and cell probe measurements allow the quantification of interactions at a single colloid/cell-level, simulating the initial attachment behavior of individual colloids/cells to surfaces. The colloidal probe technique is promising for understanding the biofouling phenomenon at the molecular level and optimizing the environmental conditions to minimize the attractive forces and maximize the repulsive forces between the foulants and substrate surfaces. Interaction force magnitudes are obtained as approach curves (attractive events or repulsive electrostatic or steric forces), retraction curves (minimum adhesive force peaks or adhesion energies calculated from the area under the x-axis), or lateral forces (if colloidal LFM is performed). Additionally, characterization of the micromechanical stiffness of abiotic surfaces and foulant layers using the colloidal probe technique provide a predictive indication of biofouling propensity, which can be used to refine the process conditions for optimal operational and cleaning efficiencies.

The colloidal probe AFM technique has proven its ability to characterize the adhesion strengths of a wide range of model biofoulants to abiotic surfaces and estimate their structural and mechanical properties, such as brush grafting density and Young's modulus. The quantification of adhesion interactions between surfaces and foulant molecules allows for the prediction of fouling behavior of such surfaces in real-time operating systems, enabling the optimization of the operational conditions that minimize biofouling. This technique, combined with other methods, can also help develop biofouling-resistant surfaces and/or surface coatings using small samples. However, it is crucial to adopt an appropriate calibration method, especially for large colloidal probes, and soft colloidal probes that cannot be conventionally calibrated using contact deflection sensitivity.

It is expected that the development of technologies that enable colloidal particles to be reversibly attached to a cantilever in a recyclable manner will expand the applications of the colloidal probe technique for use in various fields. In addition, a standardized universal procedure for calibrating colloidal probes will make the technique more routine.

Author contributions

A.E. wrote the original draft and edited the manuscript. D.J. reviewed and edited the manuscript. All authors have read and approved the submitted version of the manuscript.

List of the symbols

k	the spring constant of the cantilever	k_{TL}	the spring constant of the tipless cantilever.
E	Young's modulus of the sample	$k_{CP,c}$	the corrected colloidal probe spring constant
w	the width of the cantilever	$k_{CP,0}$	the colloidal probe spring constant calibrated by the thermal noise method
t	the thickness of the cantilever	δ_i	the correction factor of the colloidal probe cantilever
L	the length of the AFM cantilever	$\delta_{i,0}$	the standard correction pre-factor of the cantilever spring constant in the AFM software
M	the added end mass to the cantilever in the Cleveland calibration method	$\Delta G_{132}^{TOT(DLVO)}$	the total free energy of interaction according to the DLVO theory between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium)
ω	the resonance frequency of the cantilever	ΔG_{132}^{EL}	the electrostatic energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium)
m^*	the effective mass of the cantilever, which depends on the cantilever geometry	ΔG_{132}^{LW}	the Lifshitz–van der Waals energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium)
k_1	the unknown spring constant in the reference calibration method	$\Delta G_{132}^{TOT(XDLVO)}$	the total free energy of interaction according to the XDLVO theory between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium)
k_0	the spring constant of the reference cantilever	ΔG_{132}^{AB}	the Lewis acid–base energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium)
C	the nA to nm conversion factor	ΔG_{132}^{IF}	the interfacial part (IF) of the interaction energy, i.e., $(\Delta G_{132,0}^{LW} + \Delta G_{132,0}^{AB})$
D_z	the z-displacement of the sample stage	$\Delta G_{132,0}^{LW}$	the Lifshitz–van der Waals energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium) at contact ($h_0 = 0.157$ nm)
N	the deflection of the unknown cantilever after contact (nA)	$\Delta G_{132,0}^{AB}$	the Lewis acid–base energy of interaction between entities 1 (biofoulant/cell) and 2 (abiotic surface) immersed in liquid medium 3 (aqueous medium) at contact ($h_0 = 0.157$ nm)
ρ_f	the density of the fluid	γ_{ij}	the interfacial surface tension between materials i and j
Q_f	the quality factor in fluid (typically air)	γ_i^{LW}	the apolar Lifshitz–van der Waals component of the surface tension of material i
Π_i	the imaginary part of the hydrodynamic function (II) which depends only on the Reynold number ($Re = \rho_f \omega_f w^2 / 4\eta$)	γ_i^+	the polar Lewis-acid component of the surface tension of material i
ω_f	ω_f is the fundamental mode resonant frequency	γ_i^-	the polar Lewis-base component of the surface tension of material i
η	the viscosity of the fluid	$F_{\text{paraboloid}}$	the loading force of a paraboloid indenter
k_B	the Boltzmann constant	δ	the indentation depth of the sample
T	the absolute temperature	ν	the Poisson ratio of the sample
$\langle \Delta z^2 \rangle$	the mean square deflection of the cantilever due to thermal vibrations	a	the radius of the contact area
P_v	the area of the power spectrum of the thermal fluctuations alone	L_0	the thickness of the polymer brush on each side
α_i	the fraction of the total oscillation energy stored in the i-th vibration mode (α_i for the first mode of a rectangular tipless cantilever is: $\alpha_1 = 0.971$)	Γ	the number of bound polymer chains per unit area (grafting density)
χ_i	the conversion factor of the sensitivity from the end-loaded shape to the freely oscillating shape of the cantilever (≈ 1.09 for rectangular cantilevers)	D	the separation distance between the two polymer-grafted surfaces
β_i	α_i / χ_i^2	n	the mean number of segments in the polymer chains
$\hat{s}$	the correction factor that considers the condition of loading on the cantilever which accounts for the tilt of the cantilever, the torque applied on the cantilever due to the finite height of the tip, and the loading point of the cantilever, i.e., the center of the tip for standard cantilevers, and the end of the cantilever for tipless cantilevers).	l	the length of one segment in the polymer chains
S_z	the deflection sensitivity	L_0^*	the equilibrium brush thickness used in the MWC model
S_z^*	the deflection sensitivity that converts the voltage output to the effective deflection (the effective deflection is the deflection you read from the instrument after determining the sensitivity from the contact part of a force curve on a hard surface)	λ	the characteristic length or decay length of liquids (generally of water)
$P_{v,i}$	the area under the resonant peak for the i-th mode	A_{132}	the combined Hamaker constant of the biofoulant (1), abiotic surface (2), and aqueous medium (3)
ΔL	the loading offset of the cantilever tip (for sharp tips), or the loading offset of the colloidal particle (for colloidal probes)	A_{ii}	the individual Hamaker constants of material i
H	the height of the cantilever tip	l_0	the minimum equilibrium distance between two parallel flat layers of material i
θ	the tilt angle of the cantilever	ϵ_i	the static dielectric constants of material i
$k_{L-\Delta L}$	the offset-loaded spring constant of the cantilever	$\epsilon\epsilon(i\omega)$	the values of ϵ at imaginary frequencies (i ω)
k_L	the end-loaded spring constant of the cantilever	ΔG_{iwi}	the hydrophilicity/hydrophobicity of material i in water (w)
$S_{L-\Delta L}$	the offset-loaded deflection sensitivity of the cantilever	γ_{iw}	the interfacial surface tension between material i and water (w)
S_L	the end-loaded deflection sensitivity of the cantilever	γ_{iw}^{LW}	the apolar Lifshitz–van der Waals component of the interfacial surface tension between material i and water (w)
ΔL_{max}	the upper limit of the distance between the colloidal particle of diameter d and the free end of the cantilever	γ_{iw}^{AB}	the polar Lewis acid–base component of the interfacial surface
d	the diameter of the colloidal particle attached to the end of the cantilever		
k_{CP}	the spring constant of the colloidal probe corrected for the torque applied by the sphere, cantilever tilt and loading position of the sphere		
R	the radius of the attached colloidal sphere		

tension between material i and water (w)

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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